Circular Campus Requirements Report

STAGE	GROUP	COURSE	DATE
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Project

Intelligent, AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Summary

This report specifies what the Circular Campus prototype must do. KU departments use it to publish reusable assets, request and transfer them, track inspection and maintenance, and record donation, recycling or disposal. AI functions suggest classifications, rank matches, advise on sustainable actions, answer questions and draft summaries, while authorized people make every approval, release and retirement decision.

Requirement set	Count	Where
Use cases	38: 36 essential, 2 desirable (UC-33, UC-34)	Section 7
Functional requirements	121 numbered P3-FR clauses	Section 4
Non-functional requirements	30: 5 performance, 5 safety, 8 security, 12 quality	Section 5
Business rules and approval stages	16 rules, 9 approval stages	Section 3
AI and sustainability requirements	13 P3-XR requirements plus scoring and calculation rules	Section 6
Prototype 1 priorities	19 use cases, 61 FRs and 13 NFRs for the demo	Section 2
Use case relationships	4 extend and 3 include links	Section 7
Simulated stakeholder interview	16 questions mapped to requirements	Appendix A

Five decisions shape the baseline. Each physical item has its own Asset ID, so reservations, custody and history stay traceable. Custody and location change only when the receiver confirms receipt. Nobody can approve their own request or proposal. AI output is advice that a person reviews, and every AI function has a conventional fallback. Savings and carbon figures are reported only when their inputs are recorded with a source.

1 Introduction

Purpose and requirements baseline

This report specifies the required behaviour of the Intelligent AI Powered Circular Campus Resource Exchange and Asset Life Cycle Management System. It defines what the educational prototype must do, who may perform each operation, the conditions governing changes, and the evidence needed to verify the result. It is intended for the COSC 336 instructor and teaching assistant, the three student developers, and future stakeholder reviewers. Architecture and detailed implementation design belong to later phases.

The report follows the Phase 3 slides and the functional and non-functional portions of the project brief's SRS appendix [1], [2]. Its 38 detailed use cases describe separate user goals across resource registration, exchange, maintenance, final disposition, reporting, administration and AI assistance. Each has a verb-led title, explicit actors, preconditions, a trigger, exactly one main success flow, separately numbered alternatives and exceptions, and successful and unsuccessful postconditions. All requirements come first: Section 4 lists the 121 functional requirements, Section 5 the non-functional requirements and Section 6 the AI and sustainability requirements. Section 7 then gives the use-case diagram and the detailed use cases that show the behaviour behind each requirement.

The requirements are a proposed prototype baseline, not evidence that the software has been implemented or that KU has approved its operating rules. Requirements elicitation combines document analysis of the project brief with the simulated stakeholder interview in Appendix A. The interview answers are fictional and labelled as such. KU-specific procedures, role assignments, approval authority and demand remain assumptions to confirm before any live pilot.

Document conventions

P3-FR identifies a detailed functional requirement, UC identifies a use case, BR identifies a proposed business rule, P3-NFR identifies a non-functional requirement, and P3-XR identifies an additional cross-cutting requirement. A statement using “shall” is testable behaviour required by this proposed baseline. “Proposed” distinguishes a team design choice from a prescribed course requirement or verified KU practice. The older Phase 2 FR-01 to FR-06 and AI-01 to AI-05 are parent groups, not replacements for the detailed identifiers [3].

Thirty-six use cases are essential at bounded classroom-prototype depth. UC-33 Recommend a sustainable action and UC-34 Ask the resource assistant are desirable extensions because the Phase 6 minimum prototype in the brief does not list them; both remain fully specified. Essential does not mean every possible production feature is required. Desirable and future enhancements are separated in Section 2. Every numerical test threshold is a proposed acceptance target unless explicitly attributed to a source; none is a completed measurement. Section 8 explains the trace from a requirement to its use case, business rule and planned acceptance evidence.

2 System Context and Scope

Intended users and problem

The proposed system serves departments that hold reusable assets and departments that need compatible resources. A requester needs to find suitable items and track a request. A custodian needs an accurate inventory and evidence of who released and received each asset. Approvers need purpose, eligibility and ownership information before deciding. Maintenance staff need a linked inspection and repair record. Finance and sustainability staff need calculations whose inputs can be checked.

The interview scenario assumes that resource requests, approval messages, asset registers and maintenance records are held in separate emails, forms or spreadsheets. Appendix A explores how that fictional workflow creates requirements for availability, approval, custody and reporting. It does not establish KU's actual process, use of a named product, measured delays or lack of an existing workflow. Snipe-IT and Warp-it were reviewed as existing alternatives in Phase 2 [3]. Snipe-IT remains a comparison candidate, not a selected or approved implementation route. The prototype shall be implemented in Python, Java or C++ [1].

Included scope

The prototype shall support authenticated accounts, department-scoped permissions, individual asset records, publication, structured discovery, resource requests, reservations, independent approvals, collection or delivery arrangements, release, receipt and loan return. It shall retain ownership, custody, location and lifecycle history, including inspections, defects, repairs, refurbishment, donation, recycling and disposal. It shall provide notifications, audit review, conventional reports and traceable financial and environmental estimates.

All five AI functions remain included: classification and metadata suggestions; semantic matching and ranked recommendations; sustainable-action advice; natural-language assistance; and generated activity summaries. The requirements include explanations, uncertainty, human correction, version records and fallbacks. Classification covers a bounded set of text, image and document inputs rather than only asset titles. An AI recommendation does not replace a required authorization or establish a missing fact.

The evaluation environment retains Phase 2's proposed 500 synthetic assets, 5 fictional departments and 10 concurrent users [3]. Each physical item has its own Asset ID. A request for three monitors requires three distinct eligible records. The system may create a batch of records during registration, but a single record never represents unlimited stock. This per-item choice makes reservations, custody and history verifiable. It is a prototype inventory model, not a claim about KU's existing asset register.

Exclusions and priority boundaries

Live KU deployment, imports of confidential records, integration with university identity or procurement systems, public trading, payments and automatic purchasing are excluded. The prototype cannot authorize physical university transfers or disposal. Real data and any external processing require separate institutional permission. General support for every file format, unrestricted document OCR, autonomous approvals and arbitrary AI tool execution are excluded.

Priority	Included detail	Boundary
Essential prototype	UC-01 to UC-38 except UC-33 and UC-34, the classification, matching, review, configuration and summary AI functions, protected state changes, error handling, report calculations and acceptance checks	Synthetic data and bounded formats; demonstrable functions rather than screenshots standing in for implementation

Priority	Included detail	Boundary
Desirable extension	UC-33 sustainable-action advice, UC-34 conversational assistant, bulk CSV import with row errors, email notification delivery, saved searches and Arabic interface text	In-application notifications and English interaction remain the baseline; extensions cannot displace required functions
Future enhancement	Live identity and procurement integration, mobile scanning, measured demand forecasting and a multi-campus pilot	Requires new data, approvals, effort estimates and acceptance criteria

Prototype 1 priorities

Prototype 1 is the working demonstration the team will build in the six weeks before the Phase 6 deadline. Building all 38 use cases and 121 functional requirements in that time is not feasible, so Prototype 1 implements one complete end-to-end path: register and classify an asset, publish it, find it by structured and AI search, request it, reserve and approve it, release and receive it, handle a fault, and report the results. This covers the minimum prototype listed in the project brief [2]. Accounts, departments and benefit inputs are loaded as seed data for the demo instead of having their own screens.

Demo step	Prototype 1 use cases	Owner
Sign in with a role	UC-01 Sign in	Ali
Register, classify and publish	UC-04 Register an asset, UC-31 Suggest asset classifications, UC-35 Review an AI recommendation, UC-06 Publish an asset	Saeed, Humaid
Find a resource	UC-08 Search the asset catalogue, UC-32 Match resources semantically	Ali, Humaid
Request, reserve and approve	UC-10 Submit a resource request, UC-13 Reserve eligible assets, UC-14 Review a resource request	Ali, Saeed
Transfer	UC-16 Release reserved assets, UC-17 Confirm asset receipt	Saeed
Maintenance	UC-20 Report an asset fault, UC-22 Complete asset maintenance, UC-19 Record an asset inspection	Ali
Notify and report	UC-29 Read status notifications, UC-26 Generate an operational report, UC-28 Generate a sustainability report, UC-38 Summarize recorded activity	Ali, Humaid

These 19 use cases carry 61 of the 121 P3-FR clauses. Prototype 1 implements each one's main flow and the exceptions that protect data: invalid input, missing permission, unavailable quantity and AI failure. The other alternative flows follow in the later increment.

Non-functional area	Prototype 1 requirements	Why first
Performance	P3-NFR-P01 search time, P3-NFR-P02 semantic match time, P3-NFR-P05 no double reservation	These are visible in the demo and test the core exchange

Non-functional area	Prototype 1 requirements	Why first
Safety	P3-NFR-S01 unsafe assets excluded, P3-NFR-S02 inspection before return to service, P3-NFR-S03 AI never approves	Each prevents a wrong physical action
Security	P3-NFR-SEC01 server-side permission checks, P3-NFR-SEC02 hashed passwords, P3-NFR-SEC08 synthetic data only	Minimum protection for any login-based prototype
Quality	P3-NFR-Q02 keyboard access, P3-NFR-Q03 calculation fixtures, P3-NFR-Q06 AI fallback, P3-NFR-Q10 field errors	Needed for the demo to be usable and its numbers trusted

The later increment adds the remaining essential use cases: UC-02, UC-03, UC-05, UC-07, UC-09, UC-11, UC-12, UC-15, UC-18, UC-21, UC-23 to UC-25, UC-27, UC-30, UC-36 and UC-37, plus the other 17 non-functional requirements. UC-33 and UC-34 stay desirable extensions. No requirement is removed; this ordering only decides what is built and demonstrated first.

Functional ownership and actor permissions

The proposed functional owner is a KU-designated Asset Management Lead who would approve business requirements and operating rules. This names a responsibility, not a confirmed KU title or individual. The student team does not own university assets or institutional policy. A user may hold more than one assigned role, subject to separation of approval duties.

Actor	Permitted responsibility	Explicit restriction
Requester	Find permitted assets, submit and track own departmental requests, confirm receipt, report faults and initiate returns	Cannot approve own request, edit another department's inventory or receive unrelated transfers
Custodian	Register and maintain assigned inventory, publish listings, reserve units, schedule and record release, propose lifecycle actions	Cannot treat technical access as authority to approve own proposal or bypass receipt
Approver	Independently review assigned requests and lifecycle proposals, recording the decision and reason	Cannot approve an item they requested or proposed; cannot substitute for required custody confirmation
Maintenance Officer	Record inspection, fault diagnosis, maintenance actions, cost and resulting condition for assigned assets	Cannot silently retire assets or change authoritative custody during transit
Finance Officer	Validate avoided-purchase baselines and cost evidence for completed qualifying events	Cannot count an uncompleted transfer as an achieved saving
Sustainability Officer	Review reuse, repair, mass and emission-factor inputs; generate impact reports and AI summaries	Cannot turn missing input into a zero or a verified environmental claim
Administrator	Provision accounts and reference data, configure AI interfaces and restore a controlled prototype backup	Administrator role alone grants no business approval, release or retirement authority
Auditor	Read authorized events and trace changes, decisions and calculation inputs	Cannot edit operational history or inspect credentials

Department representatives use the Requester role. Procurement responsibilities are represented by the Finance Officer's validation of purchase alternatives; no purchasing transaction is implemented. An administrator managing accounts is the system administrator, not a business approver. Common account, notification and permitted AI-review functions apply to all assigned roles even where the representative actor in a use case is Requester or Custodian.

3 Business Rules Data and State Constraints

Proposed business rules

The rules below are explicit prototype operating choices. They are not verified KU policies. A functional requirement links to a rule only where it actually enforces that rule; a simple read operation does not acquire an invented business rule merely to fill a template. Shared security, audit and data constraints apply across the system.

ID	Rule
BR-01	Access is limited by assigned role, department and record relationship. Deny by default and recheck authorization on every protected operation.
BR-02	The requestor or lifecycle proposer cannot approve their own transaction. Administrator access alone provides no approval or release authority.
BR-03	Each physical item has a distinct Asset ID. Required fields must be valid, request quantities positive integers, and costs and mass nonnegative. Unknown optional values remain unknown.
BR-04	A reservable match must be permitted, listed, Available, Serviceable and compatible with every mandatory specification. Semantic similarity cannot override eligibility.
BR-05	Allocation of all requested physical IDs is atomic. An item cannot belong to two active reservations or transfers.
BR-06	Dispatch requires complete reservation, current eligibility and recorded independent approval. Material changes invalidate the affected approval and require review again.
BR-07	Authoritative custody and location change only after authorized receipt. Disputes preserve previous custody until supported resolution; an accepted damaged delivery enters UnderMaintenance, not service.
BR-08	Revision and cancellation are restricted to pre-dispatch states. They preserve history and release affected reservations consistently.
BR-09	Faulted or unsafe assets cannot be offered as working resources. Inspection evidence is required before return to service; in-transit faults never silently change custody.
BR-10	Donation, recycling and disposal require independent authorization plus execution evidence. AI cannot make these decisions.
BR-11	AI output is advice. Applying it requires human confirmation, normal eligibility checks and a trace of input, model and rules versions. Overrides record a reason.
BR-12	Avoided purchases count only for completed qualifying events with validated baselines and costs. Exclude duplicate claims and show missing inputs explicitly.
BR-13	Environmental estimates retain factor source, date, units, boundary and calculation method. Unsupported carbon estimates are unavailable, and repeated activity is not counted as distinct diverted mass.
BR-14	Operational audit events and corrections retain their history. Users cannot overwrite audit entries; logs must not expose secrets or unnecessary personal data.
BR-15	A loan preserves the original owning department. Return uses a separate transfer linked to the original loan and requires receipt by the original custodian.
BR-16	Prototype operational inputs are synthetic. Real data or transmission to external AI requires separate authorization.

Minimum information and validation

The following records describe information requirements, not database tables or an implementation schema. A stored relationship must refer to an existing authorized record. Dates use a displayed timezone and an unambiguous format; event ordering uses recorded timestamps. Required and optional fields are distinguished so the system does not invent missing acquisition or valuation facts.

Record	Required information	Validation and optional information
Asset	Asset ID, name, description, category, owning department, responsible custodian, location, condition, lifecycle state and listing flag	One physical unit per ID; acquisition date, estimated value, material, measured mass, photographs and documents may be unknown. Subcategory and technical specifications follow category definitions.
Request	Request ID, requester and department, category, purpose, mandatory specifications, quantity, preferred condition, urgency, destination, required date and transfer or loan type	Quantity must be positive; required date cannot already have passed at submission. A loan requires an expected return date after its planned receipt. Optional candidate Asset IDs express interest, not allocation.
Reservation and approval	Request ID, physical Asset IDs, reserving custodian, status, approver identity, decision, reason and time	Full requested quantity before approval; independent authorized approver; no overlapping active allocation. Approval covers the recorded request version.
Transfer	Request or original-loan reference, Asset IDs, sender and receiver, collection or delivery arrangement, planned and actual times, locations, release and receipt evidence	Release and receipt are distinct confirmations. Mismatches create a dispute. Supporting handover documents are linked to the event, not substituted for approval.
Maintenance	Asset ID, reporter, symptoms, inspection findings, assigned maintenance role, status, actions, dates, costs and post-service condition	Record actual outcomes; no negative cost. A serviceability decision needs inspection evidence. Failed repair does not make the item Available.
Lifecycle proposal	Asset IDs, proposed action, condition and rationale, proposer, reviewer, decision, execution evidence and time	Disposal is a final option requiring a reason alternatives are unsuitable. Approval and execution are separate records.
Benefit input	Completed event, avoided-purchase rationale, baseline value and evidence, repair and transfer costs, validator, date and calculation version	Missing amounts remain unavailable. Unique event references prevent the same avoided purchase being counted twice.
Environmental input	Asset ID, event, mass and its measurement or estimation basis, applicable emission factor, unit, source, version, date and system boundary	An incompatible factor or missing mass blocks that estimate. Reports distinguish measured data from modelled outcomes.
AI recommendation	Input record versions, recommendation type, model and prompt or rule versions, output, factors, uncertainty, time and review decision	Keep source record references; similarity is not a probability of correctness. Do not retain credentials in prompts, outputs or logs.
Notification and audit	Event ID, authorized recipient or actor, record reference, event type, timestamp and delivery or review status	Each committed state change creates one logical notification per intended recipient and an audit event. Retries do not duplicate the business operation.

Asset state model

Listing and condition are separate from lifecycle state. An unlisted Available asset remains in inventory but cannot be discovered as a reservable listing. Retired is terminal for operational use; correcting a mistaken historical entry requires a recorded administrative correction and new authorized review, not deletion of its history.

From	Event and guard	To and invariant
No record	Custodian saves a valid unique record	Draft; no listing or reservation
Draft	Custodian completes required fields and confirms serviceable inspection status	Available; publication requires UC-06
Available	Custodian atomically reserves the full eligible quantity for a Submitted request	Reserved; request UnderReview and reservation Active
Reserved	Independent approval, current eligibility and authorized release	InTransit; transfer Dispatched, reservation Consumed; old custody retained
Reserved	Rejection or allowed cancellation before dispatch	Available and reservation Released, provided still serviceable; otherwise UnderMaintenance
InTransit	Authorized receiver confirms every listed unit and serviceable condition	InUse; transfer Received and request Fulfilled; update custody/location and, for permanent internal transfer, owning department
InTransit	Mismatch, loss or damage is reported	InTransit and transfer Disputed; retain prior custody until resolution
InTransit	Sender and receiver resolve a dispute with evidence and maintenance disposition	Received into InUse if serviceable or UnderMaintenance if damaged and accepted; preserve incident history
InUse on loan	Borrower releases units on a linked return transfer	InTransit; original owning department unchanged
InTransit on return	Original custodian records receipt and condition	Available if serviceable, otherwise UnderMaintenance; unlisted until republished
Available or InUse	Failed inspection or confirmed fault	UnderMaintenance, unlisted; preserve custodian and location
Draft	Fault or failed inspection before publication	Draft with failed condition or NeedsInspection; publication blocked until resolved
Available	Authorized planned repair or refurbishment on an unallocated owner-held asset	UnderMaintenance and unlisted before work is scheduled; retain action authorization
Reserved	Fault makes allocation unsafe	UnderMaintenance; release affected reservation, return request to Submitted and notify for reassignment
UnderMaintenance	Completed work and recorded passed inspection	Available if held by owner, or InUse if held by borrower; no automatic publication
Eligible non-transit state	Approved donation, recycling or disposal is executed with evidence and no active allocation	Retired; retain final action, authorizations and all history

Related workflow states

Requests move from Draft to Submitted, then UnderReview when a complete reservation is established, then Approved or Rejected. A rejected request is terminal; a new request may reference it. Before dispatch an authorized cancellation produces Cancelled and releases allocation. Fulfilled means the receiver has confirmed all requested units, not merely that an approver clicked approve. Revising a Submitted request is allowed only without an active reservation; an UnderReview request must first be returned to Submitted with allocation released. Approved material changes require a fresh request version and approval.

Transfers move from Draft to Scheduled to Dispatched to Received. A pre-dispatch cancellation produces Cancelled. A receipt mismatch produces Disputed and returns to Received only after documented resolution. An unresolved dispute does not manufacture a received asset. Reservations move from Active to Consumed at release, or Released on rejection/cancellation/reassignment. There is no assumed automatic 24-hour expiry. The required date is rechecked before release; a passed date requires rescheduling and revalidation.

Maintenance moves from Open to Scheduled to InProgress to Completed, or to Cancelled with a reason. A cancelled task does not clear the underlying safety restriction. A failed repair can be recorded as Completed work with an unsuccessful outcome while the asset remains UnderMaintenance. Lifecycle proposals move from Proposed to Approved or Rejected; approved proposals become Executed only after evidence is recorded, or Cancelled with history retained. Reuse, repair and refurbishment use the normal exchange/maintenance workflows. Retention requires no custody change. Recovery of usable components is recorded through an authorized refurbishment or recycling action, not an unreviewed disappearance of the asset.

Approval stages

Every approval is a separate recorded step with its own role and decision, and no stage can be skipped. The person who requests or proposes an action cannot approve it (BR-02).

Stage	Role	Decision	Result
1 Submit request	Requester	Submit, or save as a draft	Request Submitted; no assets allocated
2 Reserve assets	Custodian	Reserve the full quantity, or report too few eligible units	Request UnderReview and units Reserved; otherwise request stays Submitted
3 Approve request	Approver (not the requester)	Approve, reject with a reason, or return for correction	Request Approved; Rejected with units released; or back to Submitted
4 Release assets	Releasing custodian	Release the approved units	Units InTransit; previous custody kept until receipt
5 Confirm receipt	Receiving requester	Confirm every unit, or report a mismatch or damage	Request Fulfilled and custody moves; or transfer Disputed
6 Propose lifecycle action	Custodian	Propose donation, recycling, disposal or another action with evidence	Lifecycle proposal Proposed
7 Authorize lifecycle action	Approver (not the proposer)	Approve or reject with a reason	Proposal Approved or Rejected
8 Execute lifecycle action	Custodian	Record execution evidence	Proposal Executed; asset Retired

Stage	Role	Decision	Result
9 Return to service	Maintenance Officer	Pass or fail the inspection	Asset Available, or stays UnderMaintenance

Cross operation safeguards

Every write shall recheck the actor, record version and permitted transition at the time of saving. Two users acting on the last available item cannot both succeed. A stale screen must show a conflict and require a refresh rather than overwrite a newer state. An interrupted transaction shall either complete the related state, custody, allocation and audit changes together or leave the prior committed state intact. A notification delivery failure may queue a retry, but must not cause a second release or receipt. These safeguards apply to ordinary forms and AI-assisted actions alike.

4 Functional Requirements

This section lists all 121 functional requirements, grouped by feature. Each P3-FR clause is numbered by the use case that exercises it, so P3-FR-10.2 is the second requirement of UC-10. The Priority column shows whether the requirement is built in Prototype 1, in the later increment, or as a desirable extension. Section 7 gives the detailed scenario behind each requirement.

4.1 Accounts and access

ID	Requirement	Use case	Priority
P3-FR-01.1	The system shall authenticate an account only when its credentials are valid and the account is enabled.	UC-01	Prototype 1
P3-FR-01.2	The system shall associate the authenticated session with the account's current role and department assignments.	UC-01	Prototype 1
P3-FR-01.3	The system shall exclude passwords from authentication responses and operational audit records.	UC-01	Prototype 1
P3-FR-02.1	The system shall restrict account creation and access changes to authorized Administrators.	UC-02	Later
P3-FR-02.2	The system shall apply revised account permissions to subsequent protected operations.	UC-02	Later
P3-FR-02.3	The system shall retain historical actor references when an account is changed or disabled.	UC-02	Later
P3-FR-37.1	The system shall restrict restoration to administrators with restoration permission.	UC-37	Later
P3-FR-37.2	The system shall block operational writes during recovery activation.	UC-37	Later
P3-FR-37.3	The system shall validate restored records and linked files before activation.	UC-37	Later
P3-FR-37.4	The system shall preserve pre-restore audit history in a protected recovery archive.	UC-37	Later

4.2 Asset registration and publication

ID	Requirement	Use case	Priority
P3-FR-03.1	The system shall restrict reference-data changes to authorized Administrators.	UC-03	Later
P3-FR-03.2	The system shall preserve identifiers referenced by existing operational records.	UC-03	Later
P3-FR-03.3	The system shall exclude inactive reference entries from new-record selections.	UC-03	Later
P3-FR-04.1	The system shall assign a distinct Asset ID to every registered physical unit.	UC-04	Prototype 1
P3-FR-04.2	The system shall reject registration values that violate mandatory-field or scope checks.	UC-04	Prototype 1
P3-FR-04.3	The system shall create registered assets in Draft state with the listing flag unset.	UC-04	Prototype 1
P3-FR-05.1	The system shall reject an asset update based on an outdated record version.	UC-05	Later
P3-FR-05.2	The system shall prevent direct edits from bypassing allocation, transfer or inspection controls.	UC-05	Later
P3-FR-05.3	The system shall preserve the previous values and correction reason for each committed asset update.	UC-05	Later
P3-FR-06.1	The system shall publish only an authorized, serviceable, unallocated asset with valid mandatory fields.	UC-06	Prototype 1
P3-FR-06.2	The system shall set a successfully published asset to Available with its listing flag set.	UC-06	Prototype 1
P3-FR-06.3	The system shall preserve custody when publishing an asset.	UC-06	Prototype 1
P3-FR-07.1	The system shall prevent listing withdrawal from silently cancelling an active allocation.	UC-07	Later
P3-FR-07.2	The system shall retain the asset record and custody after listing withdrawal.	UC-07	Later
P3-FR-07.3	The system shall exclude an unlisted asset from subsequent catalogue matching.	UC-07	Later

4.3 Search and resource requests

ID	Requirement	Use case	Priority
P3-FR-08.1	The system shall support all nine structured filters listed in step 2.	UC-08	Prototype 1
P3-FR-08.2	The system shall restrict catalogue results to eligible units meeting mandatory specifications.	UC-08	Prototype 1
P3-FR-08.3	The system shall calculate available quantity using distinct eligible physical IDs.	UC-08	Prototype 1
P3-FR-08.4	The system shall label ineligible inventory/history results not reservable.	UC-08	Prototype 1
P3-FR-09.1	The system shall check record-level access before displaying asset details.	UC-09	Later
P3-FR-09.2	The system shall display the asset's current state rather than rely on a stale search result.	UC-09	Later
P3-FR-09.3	The system shall withhold asset fields and documents outside the viewer's permission scope.	UC-09	Later
P3-FR-10.1	The system shall require a positive whole-number quantity before request submission.	UC-10	Prototype 1
P3-FR-10.2	The system shall distinguish mandatory specifications from optional request preferences.	UC-10	Prototype 1
P3-FR-10.3	The system shall create a successfully submitted request without allocating physical asset IDs.	UC-10	Prototype 1
P3-FR-11.1	The system shall save request revisions only in Submitted state without an active reservation.	UC-11	Later
P3-FR-11.2	The system shall release an UnderReview request's reservation before making its details editable.	UC-11	Later
P3-FR-11.3	The system shall retain earlier request details after a revision.	UC-11	Later

4.4 Reservation, approval and transfer

ID	Requirement	Use case	Priority
P3-FR-12.1	The system shall reject request cancellation after any linked transfer has been dispatched.	UC-12	Later
P3-FR-12.2	The system shall release the request's Active reservations atomically with pre-dispatch cancellation.	UC-12	Later
P3-FR-12.3	The system shall preserve earlier request and approval history after cancellation.	UC-12	Later
P3-FR-13.1	The system shall allocate exactly the requested quantity of distinct eligible Asset IDs.	UC-13	Prototype 1
P3-FR-13.2	The system shall commit the complete reservation and associated state changes atomically.	UC-13	Prototype 1
P3-FR-13.3	The system shall prevent reservation from creating a business approval or changing custody.	UC-13	Prototype 1
P3-FR-14.1	The system shall prevent a requester from approving their own request.	UC-14	Prototype 1
P3-FR-14.2	The system shall bind approval to the reviewed request version.	UC-14	Prototype 1
P3-FR-14.3	The system shall release reservations when a request is rejected or returned for revision.	UC-14	Prototype 1
P3-FR-15.1	The system shall associate a scheduled transfer with its approved request and reserved asset IDs.	UC-15	Later
P3-FR-15.2	The system shall preserve custody when a transfer is scheduled.	UC-15	Later
P3-FR-15.3	The system shall retain the previous schedule when a pre-dispatch schedule changes.	UC-15	Later
P3-FR-16.1	The system shall revalidate approval and asset eligibility immediately before dispatch.	UC-16	Prototype 1
P3-FR-16.2	The system shall commit dispatch for the complete reserved set atomically.	UC-16	Prototype 1
P3-FR-16.3	The system shall retain previous authoritative custody until receipt confirmation.	UC-16	Prototype 1
P3-FR-17.1	The system shall update custody only after authorized receipt confirmation.	UC-17	Prototype 1
P3-FR-17.2	The system shall preserve previous custody while a delivery dispute remains unresolved.	UC-17	Prototype 1
P3-FR-17.3	The system shall require an inspection disposition before accepting damaged units into UnderMaintenance.	UC-17	Prototype 1
P3-FR-18.1	The system shall link each return transfer to its original loan transfer.	UC-18	Later
P3-FR-18.2	The system shall preserve the original owning department throughout a loan and return.	UC-18	Later
P3-FR-18.3	The system shall leave returned assets unlisted until publication is explicitly authorized.	UC-18	Later

4.5 Maintenance and lifecycle actions

ID	Requirement	Use case	Priority
P3-FR-19.1	The system shall retain inspection findings, evidence, inspector and timestamp.	UC-19	Prototype 1
P3-FR-19.2	The system shall block service availability following a failed inspection.	UC-19	Prototype 1
P3-FR-19.3	The system shall preserve the original record when an inspection is corrected.	UC-19	Prototype 1
P3-FR-20.1	The system shall associate each fault report with one physical asset ID.	UC-20	Prototype 1
P3-FR-20.2	The system shall prevent a faulted unit from being dispatched as serviceable stock.	UC-20	Prototype 1
P3-FR-20.3	The system shall preserve authoritative custody when a fault is reported in transit.	UC-20	Prototype 1
P3-FR-21.1	The system shall link each maintenance plan to its asset and recorded maintenance need.	UC-21	Later
P3-FR-21.2	The system shall require an assigned Maintenance Officer before scheduling work.	UC-21	Later
P3-FR-21.3	The system shall preserve the asset's safety restriction when maintenance is cancelled.	UC-21	Later
P3-FR-22.1	The system shall require passing inspection evidence before restoring service after maintenance.	UC-22	Prototype 1
P3-FR-22.2	The system shall retain UnderMaintenance after an unsuccessful repair.	UC-22	Prototype 1
P3-FR-22.3	The system shall preserve the loan and owning department when a borrowed asset is repaired.	UC-22	Prototype 1
P3-FR-23.1	The system shall retain an action type, rationale and proposer for each lifecycle proposal.	UC-23	Later
P3-FR-23.2	The system shall distinguish unavailable numerical inputs from recorded zero values.	UC-23	Later
P3-FR-23.3	The system shall keep lifecycle proposals separate from execution records.	UC-23	Later
P3-FR-24.1	The system shall prevent a lifecycle proposer from approving their own proposal.	UC-24	Later
P3-FR-24.2	The system shall record the proposal version authorized by the approver.	UC-24	Later
P3-FR-24.3	The system shall require a separate execution record after lifecycle approval.	UC-24	Later
P3-FR-25.1	The system shall require approved donation, recycling or disposal authority before recording retirement.	UC-25	Later
P3-FR-25.2	The system shall require execution evidence before setting an asset to Retired.	UC-25	Later
P3-FR-25.3	The system shall prevent duplicate retirement execution records for a physical asset ID.	UC-25	Later

4.6 Reporting, notifications and audit

ID	Requirement	Use case	Priority
P3-FR-26.1	The system shall restrict report records and fields to the requesting user's permissions.	UC-26	Prototype 1
P3-FR-26.2	The system shall provide operational reports for assets, requests, transfers, maintenance, ownership changes, approvals and statuses.	UC-26	Prototype 1
P3-FR-26.3	The system shall identify the filters and data cutoff used for each generated report.	UC-26	Prototype 1
P3-FR-27.1	The system shall link each benefit assessment to its qualifying event and distinct asset IDs.	UC-27	Later
P3-FR-27.2	The system shall accept only nonnegative AED baseline values and costs.	UC-27	Later
P3-FR-27.3	The system shall exclude duplicate claims from benefit totals.	UC-27	Later
P3-FR-27.4	The system shall preserve unknown calculation inputs as unavailable rather than substitute zero.	UC-27	Later
P3-FR-28.1	The system shall derive sustainability indicators from uniquely identified qualifying events.	UC-28	Prototype 1
P3-FR-28.2	The system shall expose the source, date and unit of each environmental factor used.	UC-28	Prototype 1
P3-FR-28.3	The system shall label unsupported indicators unavailable.	UC-28	Prototype 1
P3-FR-28.4	The system shall distinguish measured inputs from estimated environmental results.	UC-28	Prototype 1
P3-FR-29.1	The system shall deliver relevant request, approval, reservation, transfer, maintenance and lifecycle status events to their designated account inboxes.	UC-29	Prototype 1
P3-FR-29.2	The system shall restrict each inbox to its addressed account.	UC-29	Prototype 1
P3-FR-29.3	The system shall recheck record access when a notification link is opened.	UC-29	Prototype 1
P3-FR-30.1	The system shall provide a scoped, chronological audit view for protected operational changes.	UC-30	Later
P3-FR-30.2	The system shall retain original audit events when corrections are recorded.	UC-30	Later
P3-FR-30.3	The system shall exclude credentials and unrestricted personal data from audit views and exports.	UC-30	Later

4.7 AI classification

ID	Requirement	Use case	Priority
P3-FR-31.1	The system shall use permitted text, JPEG/PNG image content and readable PDF text to suggest asset classifications and metadata.	UC-31	Prototype 1
P3-FR-31.2	The system shall validate suggested reference values before presenting them for acceptance.	UC-31	Prototype 1
P3-FR-31.3	The system shall retain asset fields unchanged until an authorized user confirms an update.	UC-31	Prototype 1
P3-FR-31.4	The system shall record the model, prompt or rule version associated with each classification suggestion.	UC-31	Prototype 1

4.8 AI semantic matching

ID	Requirement	Use case	Priority
P3-FR-32.1	The system shall exclude candidates that fail any mandatory eligibility condition before recommendation.	UC-32	Prototype 1
P3-FR-32.2	The system shall provide a human-readable explanation of each recommended match's score.	UC-32	Prototype 1
P3-FR-32.3	The system shall revalidate selected physical IDs during the separate reservation operation.	UC-32	Prototype 1

4.9 AI sustainable-action advice

ID	Requirement	Use case	Priority
P3-FR-33.1	The system shall explain sustainable-action recommendations using recorded evidence and disclosed uncertainty.	UC-33	Desirable
P3-FR-33.2	The system shall preserve human authorization requirements for every recommended lifecycle action.	UC-33	Desirable
P3-FR-33.3	The system shall withhold environmental estimates lacking supported calculation inputs.	UC-33	Desirable

4.10 AI resource assistant

ID	Requirement	Use case	Priority
P3-FR-34.1	The system shall restrict assistant retrieval to the requesting user's authorized scope.	UC-34	Desirable
P3-FR-34.2	The system shall identify supporting records or approved guidance for factual assistant answers.	UC-34	Desirable
P3-FR-34.3	The system shall require explicit confirmation through the normal validated workflow before an assistant-prepared action changes records.	UC-34	Desirable

4.11 AI-generated summaries

ID	Requirement	Use case	Priority
P3-FR-38.1	The system shall ground generated activity narratives in permitted recorded evidence.	UC-38	Prototype 1
P3-FR-38.2	The system shall reconcile every generated numeric assertion with deterministic report results.	UC-38	Prototype 1
P3-FR-38.3	The system shall identify generated narratives as drafts until authorized human review.	UC-38	Prototype 1
P3-FR-38.4	The system shall retain each narrative's source cutoff and model or prompt version.	UC-38	Prototype 1

4.12 AI review and configuration

ID	Requirement	Use case	Priority
P3-FR-35.1	The system shall preserve the original AI output with the attributed human review decision.	UC-35	Prototype 1
P3-FR-35.2	The system shall require a recorded reason for a human override.	UC-35	Prototype 1
P3-FR-35.3	The system shall revalidate a resulting operational action independently of AI acceptance.	UC-35	Prototype 1
P3-FR-36.1	The system shall restrict AI configuration changes to authorized administrators.	UC-36	Later
P3-FR-36.2	The system shall keep service credentials out of reports, repository content and ordinary audit text.	UC-36	Later
P3-FR-36.3	The system shall associate each AI response with its active configuration version.	UC-36	Later
P3-FR-36.4	The system shall block external AI requests without recorded input-transmission authorization.	UC-36	Later

5 Non Functional Requirements

These requirements follow SRS Sections 5.1 to 5.4 [2]. Targets retained from Phase 2 are identified below. Additional numerical limits are proposed prototype acceptance criteria selected to make testing bounded and reproducible; they are not measured performance or institutional service-level agreements. Each test record must include environment, data version, inputs, expected outcome, actual outcome and evidence. All execution results remain pending.

5 1 Performance requirements

ID	Required behaviour and pass condition	Verification owner
P3-NFR-P01	With 500 synthetic assets in 5 departments and 10 concurrent users, at least 950 of 1,000 structured searches shall return within 2 seconds after warm-up. Record workload, client-to-result latency and hardware. Retained Phase 2 target.	Humaid
P3-NFR-P02	Semantic results shall return within 5 seconds for at least 95 of 100 evaluation queries under the same dataset and concurrency, excluding one explicitly measured model warm-up. This is a proposed target, separate from structured search.	Humaid
P3-NFR-P03	Each AI generation operation shall end in a result or an explicit timeout within 30 seconds. A timeout shall preserve entered data and offer the corresponding conventional workflow. Test 10 forced slow/unavailable responses. Proposed bound.	Humaid
P3-NFR-P04	A report covering the 500-asset evaluation dataset and 1,000 synthetic lifecycle events shall finish within 5 seconds in at least 19 of 20 runs. AI narrative time is measured separately under P03. Proposed workload.	Humaid
P3-NFR-P05	In 20 paired simultaneous attempts to reserve one remaining item, exactly one attempt shall succeed per pair; no duplicate Active allocation or negative availability is permitted. Correctness takes priority over latency.	Saeed

5 2 Safety requirements

ID	Required behaviour and pass condition	Verification owner
P3-NFR-S01	The system shall exclude Unserviceable and NeedsInspection assets from working-resource allocation. All 10 negative eligibility cases shall be blocked. A warning alone is insufficient.	Ali
P3-NFR-S02	The system shall block return to service until an authorized inspection record states Serviceable. Failed repair and cancelled maintenance shall not remove the block. Test both paths and successful inspection.	Ali
P3-NFR-S03	AI suggestions shall never execute approval, release, retirement or receipt without the authorized human action and current checks. Attempt each protected transition through both direct and AI-assisted input.	Saeed
P3-NFR-S04	A disputed or damaged delivery shall not automatically mark a request Fulfilled or an asset Serviceable. Test unresolved damage, accepted damaged receipt into UnderMaintenance, and serviceable resolution.	Saeed
P3-NFR-S05	The demonstration shall identify operational records as fictional and shall not issue an authorization to use, transfer or dispose of real university property. Inspect screens, seed data and exported reports.	Ali

5 3 Security and privacy requirements

ID	Required behaviour and pass condition	Verification owner
P3-NFR-SEC01	Every protected operation shall enforce the role, department and record relationship in Section 2 on the server. All prohibited combinations in the test matrix must fail without disclosure or state change. Test direct record-ID substitution as well as interface access.	Ali
P3-NFR-SEC02	The system shall store passwords using a supported adaptive password-hashing facility, never as plaintext or reversible application encryption. Secrets shall not appear in source control, application responses, reports or audit output. Inspect configuration and attempt a seeded-secret disclosure test.	Ali
P3-NFR-SEC03	Logout and account disablement shall invalidate further protected actions for the affected session. Test browser-back, a reused request and a second active session after disablement.	Ali
P3-NFR-SEC04	Remote connections shall use encrypted transport. A local-only demonstration may use loopback, but shall not expose an unauthenticated network service. Inspect the actual deployment configuration before testing.	Saeed
P3-NFR-SEC05	The system shall accept one JPEG/PNG photograph up to 1 MB and one PDF document up to 2 MB per asset in the prototype. It shall verify actual type, restrict retrieval by permission, and reject executable or mismatched content. Unreadable scanned text shall be marked unavailable rather than invented. The 500-asset maximum is 1.5 GB of uploads before database, models and backups, consistent with the Phase 2 input allowance.	Ali
P3-NFR-SEC06	AI input shall contain only the authorized records needed for the task. Retrieved descriptions and documents shall be treated as data, not new instructions. All 5 defined prompt-injection tests must fail to reveal protected content or perform an unauthorized write. Retained Phase 2 test count.	Humaid
P3-NFR-SEC07	Ordinary users and administrators shall not edit audit events through application functions. A correction shall add a linked event, not replace the original. Test update/delete attempts and review secret redaction.	Ali
P3-NFR-SEC08	No real personal or confidential KU record shall be included in seed data or sent to an external AI service without authorization. Verify input inventories and provider settings before any live-data experiment. This is a deployment gate, not a claim of legal compliance.	Ali

OWASP's prompt-injection guidance supports separating untrusted content from instructions, validating outputs, limiting privilege and retaining human control [4]. The prototype applies these principles through SEC06 and the AI operation checks. Passing a five-case test set would show only that those defined tests passed, not that all possible attacks have been prevented.

5 4 Software quality requirements

ID	Required behaviour and pass condition	Verification owner
P3-NFR-Q01	After a 15-minute introduction, at least 4 of 5 testers shall independently search, submit a valid request and check its status. Record completion and observed errors. Retained Phase 2 usability target; testers are not described as KU staff unless that is true.	Ali
P3-NFR-Q02	Core search, request, approval and receipt tasks shall be operable by keyboard with visible focus, labelled controls and text error identification. Normal text shall meet a 4.5:1 contrast target. These selected WCAG 2.2 criteria are an acceptance subset, not a claim of full conformance [5].	Ali
P3-NFR-Q03	All 10 financial and environmental calculation fixtures in Section 8 shall reproduce expected results, units and missing-data status. Retained Phase 2 count.	Humaid
P3-NFR-Q04	Failure injection at reservation, dispatch and receipt shall leave either the complete committed operation or the prior valid state. Repeated submission of one operation shall not duplicate it. Test every affected operation at its commit boundary.	Saeed
P3-NFR-Q05	The team shall create one daily backup during active development and demonstrate one restore of records and linked files to an isolated test environment. Compare record counts, relationship integrity and attachment checksums. This is a prototype recovery requirement, not a 24-hour production guarantee.	Saeed
P3-NFR-Q06	Core conventional search, requests and reports shall remain available when an AI provider is unavailable. Ten forced AI failures shall produce explicit fallback messages, no silent fabricated AI result, and no unauthorized write.	Humaid
P3-NFR-Q07	A fresh authorized developer shall be able to install the selected prototype from the documented repository, dependency versions and seed-data instructions without shared credentials. Record one reproducible setup demonstration.	Saeed
P3-NFR-Q08	Every P3-FR and P3-XR requirement shall have a linked acceptance check. Each use case's main flow, each listed alternative and each listed exception shall have a test record before completion is claimed.	All members
P3-NFR-Q09	The selected interface shall complete the four core tasks in the two current browsers chosen for evaluation. Record browser names and versions; do not claim compatibility with untested browsers.	Ali
P3-NFR-Q10	Users shall see field-specific errors while valid entered values are retained. Test missing, malformed, out-of-range and stale values on registration, requests, approval and receipt.	Ali
P3-NFR-Q11	Every model, prompt, ranking rule and calculation change shall retain a version identifier and rerun its affected fixed evaluation cases. Results from different versions shall not be combined as one unexplained score.	Humaid
P3-NFR-Q12	Each student shall be able to explain their assigned code's inputs, outputs, rules, error handling and tests. Repository contributions and peer reviews shall retain authorship rather than treating possession of AI-generated code as demonstrated understanding.	All members

6 AI and Sustainability Measurement Requirements

Bounded AI input and output contracts

The following requirements expand cross-cutting details rather than add unconnected AI features. Model selection, framework structure and endpoint design belong to the design phase. Similarity scores must not be labelled calibrated confidence probabilities. Where no justified confidence measure exists, the interface shall state that confidence is unavailable and identify missing evidence.

ID	Requirement	Related use cases
P3-XR-01	Classification shall accept title, description and specifications and permit analysis of supported JPEG/PNG images and readable PDF text. It shall propose category, subcategory, tags, description, material and missing metadata, identify the supporting input, and leave uncertain facts for review. Unsupported scans/formats shall receive an explicit limitation.	UC-04, UC-31, UC-35
P3-XR-02	Matching shall apply permission, availability, quantity, serviceability, required-date feasibility and mandatory technical constraints before ranking. Preferred condition, purpose, category, location, urgency, transfer/repair cost and sustainability factors shall be represented in the scored comparison. No failed mandatory constraint can be compensated by another score.	UC-08, UC-13, UC-32
P3-XR-03	Each recommended match shall show a compatibility score, factor explanation, missing-input warnings, model/ranking version and the input record versions. The same authorized request shall support a comparison between structured results and AI-ranked results.	UC-32, UC-35
P3-XR-04	Sustainable-action advice shall consider continued use, internal transfer, repair, refurbishment, donation, component recovery, recycling and final disposal. It shall explain considered options using known condition, costs and safety restrictions, disclose missing evidence and require independent human authorization for action.	UC-23 to UC-25, UC-33
P3-XR-05	The assistant shall support asset discovery, request drafting/submission, history/status questions, system navigation and sustainability questions. Policy answers shall cite an approved policy record; absent one, the assistant shall say the policy is unavailable rather than invent KU rules. Writes require a displayed structured confirmation and normal authorization.	UC-34, UC-10
P3-XR-06	Generated reporting shall offer monthly activity, sustainability update, management bulletin, high-demand category, maintenance insight and recommended-action summaries from authorized query results. Every stated total shall match the supplied calculation result and period; unsupported numerical claims shall block publication.	UC-26, UC-28, UC-38
P3-XR-07	AI records shall retain source references, time, model/prompt/rule versions, output, uncertainty and human acceptance/correction/rejection. A changed asset or request version shall invalidate an unreviewed stale recommendation before application.	UC-31 to UC-36, UC-38
P3-XR-08	AI failures shall leave the ordinary classification form, structured search, human lifecycle assessment and conventional reports usable. A fallback shall be labelled conventional; it does not count as successful implementation of the AI feature.	UC-31 to UC-38

ID	Requirement	Related use cases
P3-XR-13	Structured discovery shall provide category, condition, department, location, minimum eligible quantity, availability, asset age, estimated value and lifecycle-status filters. The reservable catalogue shall include only eligible listed units. A separately authorized inventory/history mode may show other states, clearly marked not reservable. Unknown acquisition dates or values shall not pass a numeric range filter as zero.	UC-08, UC-09, UC-26

Proposed ranking interpretation

After mandatory filters pass, the proposed prototype compares feasible alternatives using five normalized components: semantic and functional suitability 0.45, technical/preferred-condition fit 0.20, date and location readiness relative to urgency 0.15, combined repair and transfer cost 0.10, and supported environmental benefit 0.10. Compatibility score equals 100 multiplied by the weighted sum. These weights are an explicit starting hypothesis for evaluation, not trained findings or KU policy. The ranking configuration shall record its version and rationale.

Each component lies between 0 and 1 under a documented rubric. Suitability compares the expressed purpose; preferred-condition fit never relaxes mandatory safety; readiness reflects whether the estimated transfer date meets the request's required date, with urgency determining how much schedule slack is acceptable. Cost prefers lower documented combined cost within the candidate set; equal known costs produce equal scores. Sustainability uses comparable waste-diversion or emission estimates within the same stated boundary. Request urgency is constant within a single request and must not be presented as a differentiating factor unless it changes candidate-specific readiness.

If a nonmandatory factor is missing for any candidate in a compared set, exclude that factor for all candidates in that comparison and renormalize the remaining weights. Display the omission so missing cost or environmental data never becomes a free advantage. Do not combine incomparable carbon estimates. If mandatory specifications or date feasibility cannot be established, flag the item for clarification rather than present it as eligible. Equal scores use a disclosed stable Asset ID tie-break, not a department preference. The weights, rubrics and missing-data rule must be evaluated together before the design baseline is approved.

Financial and environmental definitions

ID	Measure and required calculation	Counting control
P3-XR-09	Avoided-purchase benefit per qualifying event equals supported replacement-purchase baseline minus repair cost minus transfer cost. Preserve negative results rather than silently converting losses into savings.	Include only a completed event with finance-validated rationale; no claim simply because an item was listed or approved.
P3-XR-10	Reuse count equals distinct completed qualifying reuse events in the stated period. Proposed cohort reuse rate equals 100 multiplied by the number of distinct Asset IDs first published for reuse in the selected period that complete qualifying reuse by the report cutoff, divided by all distinct Asset IDs first published for reuse in that period. Reused mass counts distinct qualifying assets once per period; repair count and recycling mass remain separate.	Record the cohort period and report cutoff. A zero denominator makes the rate unavailable. Disclose elapsed follow-up and cohort maturity: a recently published cohort has had less opportunity for reuse, so its early rate is not a final outcome or directly comparable with a longer-followed cohort. A loan return or repeated export creates no additional avoided purchase. Do not add overlapping reuse, repair and recycling categories into an unduplicated mass total.
P3-XR-11	Estimated waste diversion includes only recorded reuse or recycling with a stated disposal counterfactual. Asset-life extension equals observed continued service after the documented expected retirement date, or a separately labelled scenario when future service is assumed.	No retirement baseline means life extension is unavailable. Measured mass alone does not prove that disposal was avoided.
P3-XR-12	Estimated carbon reduction equals supported avoided-production/disposal emissions minus supported additional repair and transfer emissions within a consistent boundary. Emission factors shall come from the U.S. EPA Waste Reduction Model (WARM) [6], recording the WARM version, material category, unit and limitations for each factor.	Missing or incompatible factors produce an unavailable estimate. No default “carbon saved per asset” is permitted.

The figures below are synthetic test values chosen by the team to exercise the calculations; they are not market prices or KU records. They assume AED 800 of avoided purchasing cost minus AED 150 repair and AED 50 transfer cost, giving AED 600 net benefit per qualifying event, as used in the Phase 2 scenario [3]. Real baselines must come from finance-validated purchase records under P3-XR-09. Twenty such events would yield AED 12,000 net avoided purchasing cost before project costs; subtracting the Phase 2 first-year economic cost of AED 9,120 gives AED 2,880. These values motivate traceable input and report requirements, not a claim of achieved KU savings. The prototype shall store the inputs needed to replace these assumptions with authorized evidence.

Conventional and AI comparison

Task	Conventional behaviour	AI addition and evaluation boundary
Classification	Custodian selects a defined category and fills validated fields	AI suggests labels and missing fields; assess accuracy against labelled examples and retain corrections
Resource discovery	Exact filters and deterministic eligibility checks	Semantic comparison may retrieve equivalent descriptions; compare relevance on the same fixed request set without relaxing constraints
Lifecycle decision	Authorized people inspect facts and follow business rules	AI compares supported options and explains tradeoffs; no autonomous approval or claimed expert safety certification
Resource interaction	User completes forms and follows status links	Assistant interprets natural language and proposes structured actions; confirmation and permissions remain unchanged
Reporting	Queries and verified formulas generate totals	AI explains supplied totals and trends; every number remains reconcilable to the conventional result

7 Use Cases and Diagrams

Catalogue and diagram conventions

The catalogue identifies the 38 user goals, their Phase 2 parent groups and proposed verification owners. The functional requirements for each use case are listed in Section 4. The goal statement provides the user-level requirement, and the numbered “shall” clauses provide the developer and tester-level requirements. The main flow describes actor inputs and observable system responses. Alternative and exception identifiers include the originating main-flow step and specify whether processing rejoins or ends.

The overall diagram contains one system boundary and all 38 use cases. It emphasizes initiating actor associations; supporting interactions and shared-role access are specified in the detailed cases and actor table. Repeated instances of a named actor denote the same role, not extra roles. Actor associations do not indicate chronological sequence or confer approval authority. Internal databases and AI components are not external actors. For a separately hosted AI service, the service is a supporting external interface within the AI requirements, not an independent business decision maker.

Use case	User goal	Parent group	Proposed owner
UC-01	Sign in	FR-01	Ali
UC-02	Manage user access	FR-01	Ali
UC-03	Maintain reference data	FR-02	Ali
UC-04	Register an asset	FR-02	Saeed
UC-05	Update an asset record	FR-02	Saeed
UC-06	Publish an asset	FR-02	Saeed
UC-07	Withdraw an asset listing	FR-02	Saeed
UC-08	Search the asset catalogue	FR-03	Ali
UC-09	Inspect asset details	FR-03	Ali
UC-10	Submit a resource request	FR-03	Ali
UC-11	Revise a pending request	FR-03	Ali
UC-12	Cancel a resource request	FR-04	Saeed
UC-13	Reserve eligible assets	FR-04	Saeed
UC-14	Review a resource request	FR-04	Saeed
UC-15	Schedule an asset transfer	FR-04	Saeed
UC-16	Release reserved assets	FR-04	Saeed
UC-17	Confirm asset receipt	FR-04	Saeed
UC-18	Return loaned assets	FR-04	Saeed
UC-19	Record an asset inspection	FR-05	Ali
UC-20	Report an asset fault	FR-05	Ali
UC-21	Plan asset maintenance	FR-05	Ali

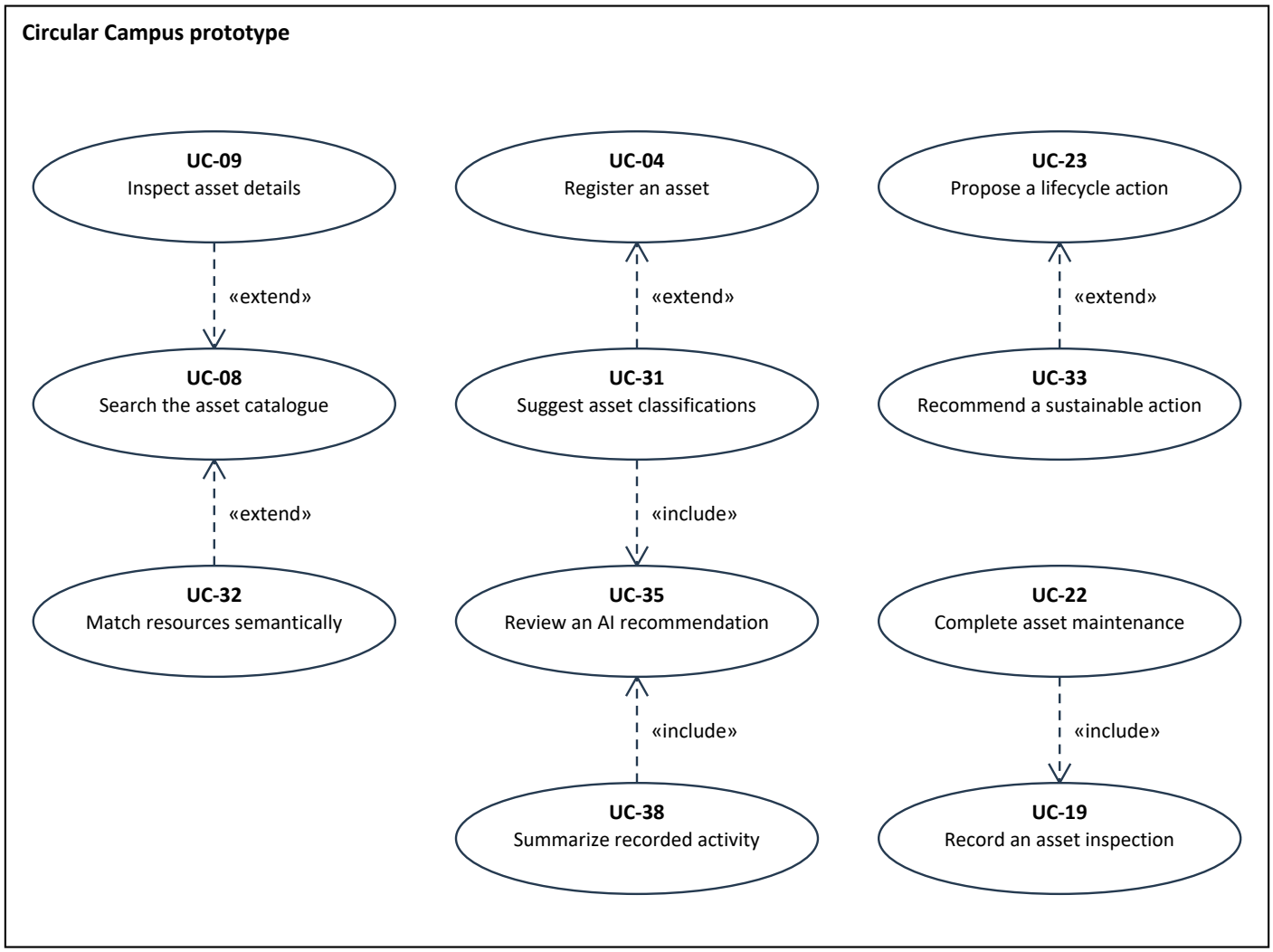
Use case	User goal	Parent group	Proposed owner
UC-22	Complete asset maintenance	FR-05	Ali
UC-23	Propose a lifecycle action	FR-05	Humaid
UC-24	Authorize a lifecycle action	FR-05	Humaid
UC-25	Record asset retirement	FR-05	Humaid
UC-26	Generate an operational report	FR-06	Humaid
UC-27	Record benefit calculation inputs	FR-06	Humaid
UC-28	Generate a sustainability report	FR-06	Humaid
UC-29	Read status notifications	FR-06	Ali
UC-30	Review the audit history	FR-06	Ali
UC-31	Suggest asset classifications	AI-01	Humaid
UC-32	Match resources semantically	AI-02	Humaid
UC-33	Recommend a sustainable action	AI-03	Humaid
UC-34	Ask the resource assistant	AI-04	Humaid
UC-35	Review an AI recommendation	AI-01 to AI-05	Humaid
UC-36	Configure AI services	AI-01 to AI-05	Saeed
UC-37	Restore a prototype backup	FR-01	Saeed
UC-38	Summarize recorded activity	AI-05	Humaid

Use case relationships

Seven relationships connect use cases where one goal always reuses another or optionally adds to it. An «include» arrow runs from the base use case to the behaviour it always performs. An «extend» arrow runs from the optional use case to the base use case it adds to.

Relationship	Meaning in this system
UC-31 «include» UC-35	Every classification suggestion goes through human review before any field changes
UC-38 «include» UC-35	Every generated summary is reviewed before it is saved
UC-22 «include» UC-19	Completing maintenance always records a passing inspection
UC-09 «extend» UC-08	A requester may open an asset's details from search results
UC-32 «extend» UC-08	A requester may add AI semantic ranking to a structured search
UC-31 «extend» UC-04	A custodian may ask for AI classification while registering an asset
UC-33 «extend» UC-23	A custodian may ask for AI sustainable-action advice while proposing a lifecycle action

Sign in is a precondition of the other use cases, not an included use case, because it happens before them rather than inside their flows.

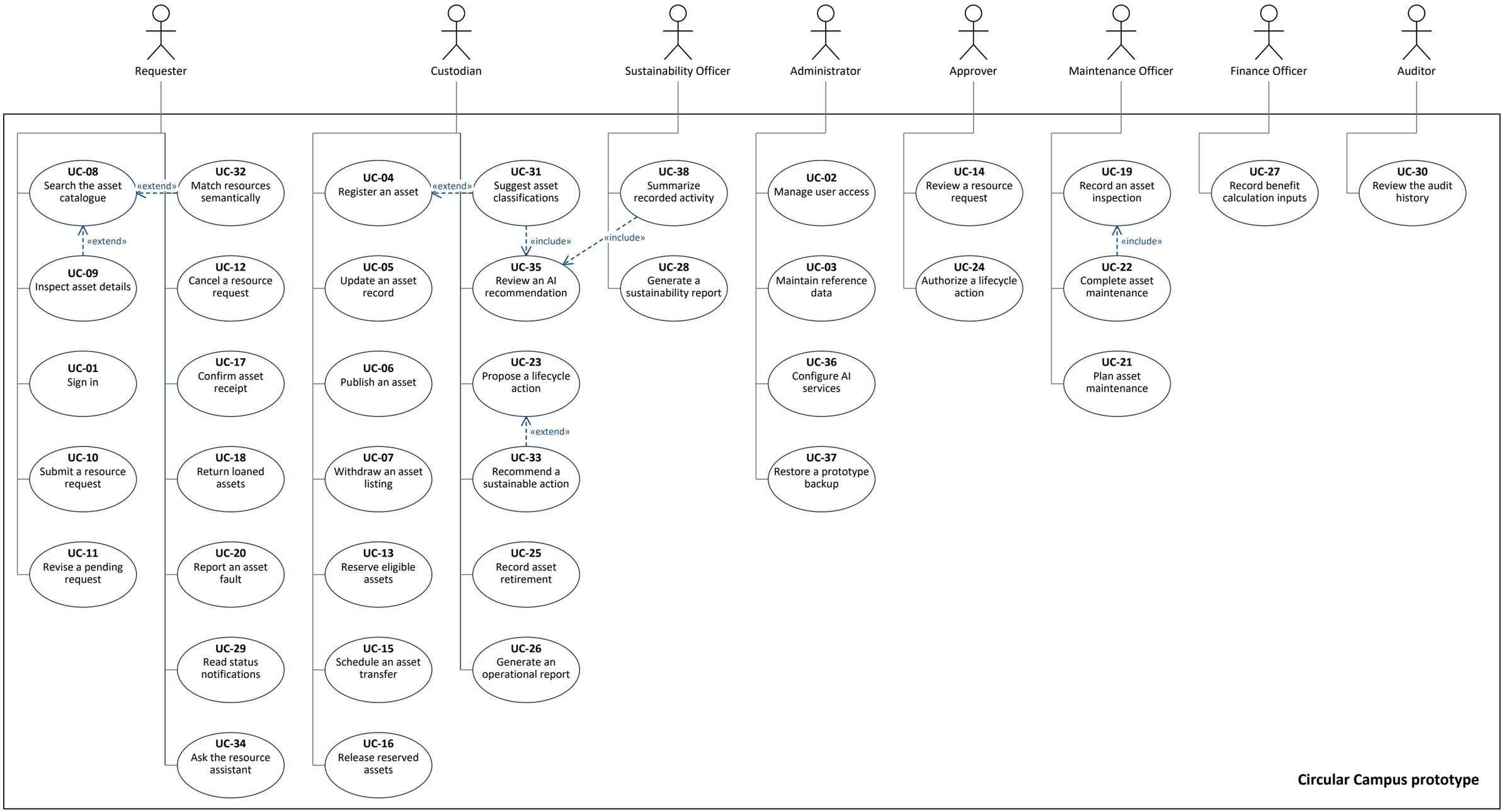


Detailed use cases

The following use cases are the normative behavioural detail for this proposed baseline. The preconditions describe facts that must hold before the trigger. Validation of user-entered values occurs in the flow rather than being hidden as a precondition. Unless a more specific exception states otherwise, a failed operation preserves the last committed business state and records a safe error without exposing secrets. A successful read does not change an asset's state. Test examples use synthetic records only.

Circular Campus overall use case diagram

All 38 goals within one system boundary. Solid lines show initiating actor associations; dashed arrows show include and extend relationships.



Each actor appears once; the Requester and Custodian associations branch to two columns of use cases. Common account and inbox access applies to every role.

This is a use case map, not a sequence or an approval hierarchy. The data store and internal AI components are inside the system.

UC-01 Sign in

Priority: Essential, Prototype 1. Parent group: FR-01. Proposed verification owner: Ali Alawadhi.

Goal: Establish an authenticated session with the permissions assigned to an existing prototype account, without granting additional business authority.

Primary actor: Requester (representative of every account role). Supporting actors: None.

Preconditions: P1. An account has been provisioned in the prototype. P2. The user can access the sign-in interface without an authenticated session.

Trigger: The user chooses to sign in to the prototype.

Main flow

1. The system presents the account-identifier and password fields.
2. The user enters the account identifier and password and submits the form.
3. The system validates the submitted credentials and confirms that the account is enabled.
4. The system loads the account's current roles, department assignments and permitted record scope.
5. The system establishes an authenticated session without including the password in its response or operational audit.
6. The system records the authentication outcome and opens the user's permitted starting view. Subsequent protected actions remain subject to fresh authorization checks.

Alternative flows

A1 at main step 4. The account holds more than one assigned role. The system presents the functions available through those assignments. The user selects a permitted working view; selecting it does not add permissions or remove the prohibition on approving the user's own proposals. Return to main step 5.

Exception flows

E1 at main step 3. Credentials are incorrect or the account is disabled. The system displays a generic sign-in failure without revealing whether the identifier exists. The system retains no authenticated session from this attempt and does not expose credentials in its failure record. End unsuccessfully; the user may start another sign-in attempt.

Success postconditions: The session identifies the enabled account and its current permitted scope.

Unsuccessful postconditions: The attempt grants no access and changes no account assignments.

Applicable business rules: BR-01, BR-02, BR-14.

Functional requirements: P3-FR-01.1, P3-FR-01.2, P3-FR-01.3 (listed in Section 4).

Acceptance check: A valid enabled synthetic Requester reaches the requester view. The same credentials fail after deactivation, and an incorrect password creates no authenticated session or credential-bearing log entry.

UC-02 Manage user access

Priority: Essential, later increment. Parent group: FR-01. Proposed verification owner: Ali Alawadhi.

Goal: Maintain prototype account access so assigned roles and departments determine subsequent protected actions without altering historical ownership of work.

Primary actor: Administrator. Supporting actors: None.

Preconditions: P1. The Administrator is authenticated. P2. The reference roles and departments exist in the prototype.

Trigger: The Administrator opens access management to change an account's assignments or enabled status.

Main flow

1. The system verifies the Administrator's permission to manage accounts.
2. The Administrator selects the target account, and the system displays its current enabled status, roles and department assignments.
3. The Administrator selects the intended assignments or deactivation and submits the change.
4. The system validates the referenced roles and departments and rechecks the Administrator's authority.
5. The system saves the access change and appends an audit event identifying the administrator, target account, time and changed assignments.
6. The system confirms the result. Subsequent protected operations use the revised permissions, while earlier requests, approvals and audit events retain their original actor references.

Alternative flows

A1 at main step 2. A new synthetic account is required. The Administrator selects account creation and supplies a unique identifier, initial access details and proposed assignments. The system presents the new account configuration for confirmation. Return to main step 3; successful completion creates the account.

Exception flows

E1 at main step 4. An assignment references an unavailable department, or the acting account no longer has administration permission. The system rejects the submitted change and identifies the permitted correction without exposing restricted account data. Existing access assignments remain unchanged. End unsuccessfully without an access change.

Success postconditions: The recorded access configuration governs the target account's next protected operation.

Unsuccessful postconditions: No partial assignment or account deactivation is committed.

Applicable business rules: BR-01, BR-02, BR-14, BR-16.

Functional requirements: P3-FR-02.1, P3-FR-02.2, P3-FR-02.3 (listed in Section 4).

Acceptance check: Remove a Custodian assignment and verify that the account's next asset update is denied. An Administrator-only account must still be denied a transfer approval, and a non-Administrator access change must leave assignments intact.

UC-03 Maintain reference data

Priority: Essential, later increment. Parent group: FR-02. Proposed verification owner: Ali Alawadhi.

Goal: Maintain the prototype's department, location and category choices while preserving the meaning of records that already use them.

Primary actor: Administrator. Supporting actors: None.

Preconditions: P1. The Administrator is authenticated. P2. Reference-data management is available in the configured prototype.

Trigger: The Administrator requests a new reference entry or a change to an existing entry.

Main flow

1. The system checks the Administrator's reference-data permission and displays the selected reference group.
2. The Administrator selects an entry to edit, and the system displays its identifier, current values and active status.
3. The Administrator enters the revised label or permitted reference attributes and submits the change.
4. The system validates mandatory fields and linked references, then checks whether the change would break an existing asset or request reference.
5. The system saves the valid change against the existing reference identifier and records the old and new values in the audit history.
6. The system displays the updated entry and makes its active values available in subsequent forms without rewriting historical operational events.

Alternative flows

A1 at main step 2. The Administrator chooses to add an entry instead of editing one. The system presents an empty entry form and allocates an identifier when the validated entry is saved. Return to main step 3.

A2 at main step 3. An entry should no longer be selected in new records. The Administrator requests deactivation rather than deletion. The system prepares to retain existing links and exclude the entry from new selections. Return to main step 4 to validate the proposed deactivation.

Exception flows

E1 at main step 4. A required field is blank or the proposed change would leave a broken reference. The system identifies the invalid field or dependency and rejects the change. Return to main step 3; no reference change is saved.

Success postconditions: The reference entry is usable as configured, and earlier linked records remain interpretable.

Unsuccessful postconditions: Existing reference values and dependent records remain unchanged.

Applicable business rules: BR-01, BR-03, BR-14.

Functional requirements: P3-FR-03.1, P3-FR-03.2, P3-FR-03.3 (listed in Section 4).

Acceptance check: Deactivate a category used by a synthetic asset: its historical association remains visible, but a new registration cannot select it. A blank replacement label is rejected without changing the existing entry.

UC-04 Register an asset

Priority: Essential, Prototype 1. Parent group: FR-02. Proposed verification owner: Saeed Alkaabi.

Goal: Create a traceable record for an individual synthetic physical asset without publishing it or making it automatically requestable.

Primary actor: Custodian. Supporting actors: None.

Relationships: This use case is extended by UC-31 Suggest asset classifications.

Preconditions: P1. The Custodian is authenticated and assigned to a department. P2. The necessary department, location and category references exist.

Trigger: The Custodian selects asset registration.

Main flow

1. The system verifies registration permission and presents the asset form with permitted reference choices.
2. The Custodian enters the asset name, category, owning department, custodian, location, condition and available specifications, identifying the physical unit being recorded.
3. The Custodian supplies any available optional purchase, value or supporting information and submits the registration.
4. The system validates mandatory fields, allowed assignments and nonnegative recorded monetary or mass values; omitted optional values remain unavailable rather than becoming zero.
5. For each physical unit, the system allocates a distinct Asset ID and saves a Draft with its listing flag unset.
6. The system appends the registration event and displays the new identifier and recorded details. Publication requires the separate publishing use case.

Alternative flows

A1 at main step 2. The Custodian registers several equivalent physical units together. The Custodian enters a positive whole-number count and the identifying details required for each unit. The system prepares that number of individual records, each requiring its own distinct Asset ID; a shared description does not create pooled stock under one ID. Return to main step 3 for the batch submission.

Exception flows

E1 at main step 4. A mandatory field is missing, an assignment is outside scope, or a recorded amount is negative. The system highlights the failed checks and saves no invalid asset record. The Custodian corrects the submitted values. Return to main step 3.

Success postconditions: Each registered physical unit has a distinct Draft record and remains unlisted.

Unsuccessful postconditions: Invalid registration creates no asset record and overwrites no existing asset.

Applicable business rules: BR-01, BR-03, BR-14, BR-16.

Functional requirements: P3-FR-04.1, P3-FR-04.2, P3-FR-04.3 (listed in Section 4).

Acceptance check: Register three synthetic monitors in a batch and obtain three distinct unlisted Draft IDs. Submitting a negative value or a missing department creates no asset record.

UC-05 Update an asset record

Priority: Essential, later increment. Parent group: FR-02. Proposed verification owner: Saeed Alkaabi.

Goal: Correct permitted asset information while protecting allocations, authoritative custody and the historical record.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The Custodian is authenticated. P2. The asset record exists within the Custodian's assigned record scope.

Trigger: The Custodian opens an asset for editing.

Main flow

1. The system checks edit permission and displays the asset's current values, state and version.
2. The Custodian changes permitted descriptive information, specifications or recorded values and enters the reason for correction.
3. The system validates required fields and nonnegative amounts and determines whether the proposed change affects an allocation or protected life-cycle field.
4. The system rechecks the record version, current allocation status and the Custodian's permission before accepting the update.
5. The system saves the permitted changes without replacing the Asset ID or earlier operational events.
6. The system appends the actor, time, correction reason and old and new values, then displays the updated record.

Alternative flows

A1 at main step 3. The proposed correction changes an allocated asset's material specifications or authoritative custody, ownership, location or condition. The system does not apply the direct edit. It identifies the relevant request-revision, cancellation, transfer or inspection workflow. Any permitted material change must follow that workflow, including invalidating affected approval and releasing reservations where required. End without the direct update; restart after the relevant workflow establishes an editable state.

Exception flows

E1 at main step 4. Another operation has changed the asset version or reserved it since the form was opened. The system rejects the stale submission and displays the current permitted values. The Custodian reviews the differences instead of overwriting the concurrent change. Return to main step 2.

Success postconditions: The permitted correction is saved with its change history; custody and allocation remain consistent.

Unsuccessful postconditions: The attempted edit changes neither authoritative custody nor a concurrent reservation.

Applicable business rules: BR-01, BR-03, BR-06, BR-07, BR-09, BR-14.

Functional requirements: P3-FR-05.1, P3-FR-05.2, P3-FR-05.3 (listed in Section 4).

Acceptance check: Correct an unallocated monitor's specification and retain both versions. If another session reserves that unit before saving, the stale edit fails and neither the reservation nor custody is overwritten.

UC-06 Publish an asset

Priority: Essential, Prototype 1. Parent group: FR-02. Proposed verification owner: Saeed Alkaabi.

Goal: Make a permitted, serviceable physical asset discoverable and eligible for a future reservation without allocating it to anyone.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The Custodian is authenticated. P2. The asset record exists within the Custodian's assigned scope.

Trigger: The Custodian requests publication of the asset.

Main flow

1. The system retrieves the current asset record and checks the Custodian's publication permission.
2. The system presents the proposed listing, including its Asset ID, specifications, condition, department and location.
3. The Custodian reviews the listing and confirms publication.
4. The system verifies mandatory publication fields, Serviceable condition, Draft or Available state, absence of active allocation and any required return-to-service inspection evidence.
5. The system atomically sets the asset to Available and its listing flag to listed, preserving its owning department and custody.
6. The system records the publication event and confirms that the asset can appear in permitted catalogue searches. A reservation still requires a separate successful allocation.

Alternative flows

A1 at main step 2. The asset is already listed and remains eligible. The system displays its existing publication status and current listing. It treats the repeated publication request as having no additional state effect. End successfully with the original listing retained and no duplicate asset.

A2 at main step 3. The Custodian identifies an incorrect description before confirmation. The Custodian leaves publication and corrects the record through UC-05. End without publication; restart UC-06 after the correction.

Exception flows

E1 at main step 4. The asset is unserviceable, lacks a required field, or has become reserved. The system states the blocking condition and declines publication. End unsuccessfully without changing state, listing, allocation or custody.

Success postconditions: The single physical unit is Available and listed, but not reserved.

Unsuccessful postconditions: The existing asset and allocation states remain unchanged.

Applicable business rules: BR-01, BR-03, BR-04, BR-05, BR-09, BR-14.

Functional requirements: P3-FR-06.1, P3-FR-06.2, P3-FR-06.3 (listed in Section 4).

Acceptance check: Publish a complete Serviceable Draft monitor and find its ID in an authorized search. Publication of an Unserviceable unit fails; repeating a successful publication does not create another unit.

UC-07 Withdraw an asset listing

Priority: Essential, later increment. Parent group: FR-02. Proposed verification owner: Saeed Alkaabi.

Goal: Remove an unallocated asset from discovery without deleting the physical asset record, its custody or its operational history.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The Custodian is authenticated and assigned to the asset. P2. The asset is listed, Available and has no active reservation when the withdrawal is initiated.

Trigger: The Custodian chooses to withdraw the listing.

Main flow

1. The system retrieves the asset and checks the Custodian's permission to withdraw its listing.
2. The system displays the Asset ID and explains that withdrawal removes discovery availability but does not retire or delete the asset.
3. The Custodian supplies the withdrawal reason and confirms the action.
4. The system rechecks the record version and absence of an active reservation or transfer before changing the listing flag.
5. The system sets the listing flag to unlisted while retaining Available state, the owning department, current custody and existing history.
6. The system records the actor, time and withdrawal reason and confirms removal from subsequent catalogue results and candidate allocations.

Alternative flows

A1 at main step 3. The Custodian decides the item should remain discoverable. The Custodian cancels the withdrawal before confirmation. The system returns to the unchanged asset record. End without withdrawing the listing.

A2 at main step 4. Another authorized operation has already withdrawn the same listing without allocating the asset. The system displays the current unlisted status and retains the original withdrawal event. End successfully without another state change.

Exception flows

E1 at main step 4. A concurrent reservation or transfer now exists. The system rejects the direct withdrawal and preserves the allocation. It directs the Custodian to the applicable pre-dispatch request or transfer workflow rather than silently removing the reserved asset. End unsuccessfully; allocation remains authoritative.

Success postconditions: The asset is unlisted but remains recorded, Available and traceable.

Unsuccessful postconditions: No active allocation, custody value or asset history is lost.

Applicable business rules: BR-01, BR-04, BR-05, BR-08, BR-14.

Functional requirements: P3-FR-07.1, P3-FR-07.2, P3-FR-07.3 (listed in Section 4).

Acceptance check: Withdraw one listed synthetic chair: catalogue search omits it while its ID and history remain accessible to the authorized Custodian. A reservation committed before withdrawal causes the withdrawal to fail without releasing that reservation.

UC-08 Search the asset catalogue

Priority: Essential, Prototype 1. Parent group: FR-03. Proposed verification owner: Ali Alawadhi.

Goal: Find eligible physical units or inspect authorized inventory using explicit structured filters.

Primary actor: Requester. Supporting actors: None.

Relationships: This use case is extended by UC-09 Inspect asset details; is extended by UC-32 Match resources semantically.

Preconditions: P1. The Requester has an authenticated prototype account.

Trigger: The Requester opens catalogue search.

Main flow

1. The system displays catalogue mode, permitted reference choices and search controls.
2. The Requester submits the selected mode, keywords, mandatory specifications and filters: category, condition, department, location, quantity, availability, asset age, value and lifecycle status.
3. The system validates supplied ranges, references and positive integer quantity, then applies current access scope.
4. The system selects listed, Available, Serviceable units meeting mandatory specifications. Quantity n requires at least n distinct eligible IDs in a matching category/specification group.
5. The system displays individual IDs, current states and permitted details; group counts do not turn one asset record into pooled stock.
6. The Requester reviews results. Searching changes no request, reservation or asset state; detailed inspection follows UC-09.

Alternative flows

A1 at main step 2. No filters are supplied. The system browses the permitted catalogue. Return to main step 3.

A2 at main step 4. Authorized inventory/history mode was selected. The system applies filters across all permitted states, retaining the eligible-ID quantity calculation. It labels ineligible units not reservable. Return to main step 5; viewing does not grant allocation rights.

A3 at main step 4. No results satisfy the filters. The system shows an empty result without relaxing mandatory constraints. End with no matches.

Exception flows

E1 at main step 3. A filter is invalid or the selected mode is unauthorized. The system rejects the query without broadening access. Return to main step 2.

Success postconditions: Results reflect the authorized mode, filters and current eligibility.

Unsuccessful postconditions: No record is modified and no restricted listing is disclosed.

Applicable business rules: BR-01, BR-04.

Functional requirements: P3-FR-08.1, P3-FR-08.2, P3-FR-08.3, P3-FR-08.4 (listed in Section 4).

Acceptance check: Quantity two excludes a matching group with one eligible ID and one Reserved ID. With quantity cleared, authorized history search can show the Reserved unit as not reservable; catalogue search excludes it. A reversed age range fails validation.

UC-09 Inspect asset details

Priority: Essential, later increment. Parent group: FR-03. Proposed verification owner: Ali Alawadhi.

Goal: Read the current permitted information for a physical unit and understand whether it can satisfy a resource need.

Primary actor: Requester. Supporting actors: None.

Relationships: This use case extends UC-08 Search the asset catalogue.

Preconditions: P1. The Requester is authenticated. P2. The Requester has an Asset ID from a catalogue result or a permitted linked request.

Trigger: The Requester opens the asset's details.

Main flow

1. The system receives the selected Asset ID and retrieves the current record.
2. The system checks whether the Requester may view that record and which fields or linked documents are within the account's scope.
3. The system presents the identifier, name, category, specifications, condition, current state and permitted department and location information.
4. The system displays permitted supporting information and whether the asset currently meets catalogue eligibility; viewing does not guarantee a later reservation.
5. The Requester reads the details, completing the inspection. Resource-request submission is a separate UC-10 action; viewing changes no operational state.

Alternative flows

A1 at main step 3. The asset is no longer Available but the Requester retains access through an existing request. The system shows its current state rather than the previous search result's status. The system does not offer the unit as newly available or reveal unrelated requests associated with it. Return to main step 4.

A2 at main step 4. An optional specification or supporting document was not recorded. The system identifies the information as unavailable without inventing a value or interpreting absence as a favourable condition. Return to main step 5.

Exception flows

E1 at main step 2. The Asset ID is inaccessible to this account or no longer resolves to a viewable record. The system declines to disclose the record and returns a neutral unavailable response. End unsuccessfully without exposing restricted fields.

Success postconditions: The Requester sees current information within their access rights.

Unsuccessful postconditions: No asset state changes and no unauthorized record information is revealed.

Applicable business rules: BR-01, BR-04.

Functional requirements: P3-FR-09.1, P3-FR-09.2, P3-FR-09.3 (listed in Section 4).

Acceptance check: After a searched asset becomes Reserved, an authorized linked Requester sees Reserved rather than Available. A direct request for a prohibited asset ID exposes neither its details nor its restricted attachments.

UC-10 Submit a resource request

Priority: Essential, Prototype 1. Parent group: FR-03. Proposed verification owner: Ali Alawadhi.

Goal: Record a departmental resource need with explicit quantity and compatibility requirements for subsequent custodian review.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The Requester is authenticated and assigned to a requesting department. P2. The category and department references used by request forms exist.

Trigger: The Requester chooses to create a resource request.

Main flow

1. The system checks request-creation permission and displays the request form with permitted department choices.
2. The Requester enters the purpose, category, mandatory specifications, positive whole-number quantity, destination, required date and whether the need is a loan or permanent internal transfer.
3. The Requester separates optional preferences from mandatory constraints and confirms submission.
4. The system validates mandatory fields, quantity, date format and departmental authority without treating a preferred candidate as guaranteed stock.
5. The system creates the request in Submitted state with a distinct Request ID and no active reservation or approval.
6. The system records submission, makes the request available to the permitted review workflow and shows the Requester its identifier and current status.

Alternative flows

A1 at main step 3. The Requester wants to retain incomplete work without submitting it. The Requester chooses to save a draft. The system records a Draft visible within the Requester's permitted scope and does not place it in the submitted review workflow. End with a saved Draft, not a Submitted request.

Exception flows

E1 at main step 4. Quantity is zero, negative or fractional, or another mandatory value is invalid. The system identifies the failed validation and retains the entered form for correction. No Submitted request or reservation is created. Return to main step 2.

Success postconditions: The Submitted request defines its full need and awaits review without allocating assets.

Unsuccessful postconditions: Invalid input does not create a Submitted request or alter asset availability.

Applicable business rules: BR-01, BR-03, BR-05, BR-14, BR-15.

Functional requirements: P3-FR-10.1, P3-FR-10.2, P3-FR-10.3 (listed in Section 4).

Acceptance check: Submit a synthetic two-monitor loan request and verify Submitted status with zero reserved units. A quantity of 1.5 is rejected, and saving incomplete work creates only a Draft.

UC-11 Revise a pending request

Priority: Essential, later increment. Parent group: FR-03. Proposed verification owner: Ali Alawadhi.

Goal: Correct a pending need without allowing an earlier allocation or review to authorize changed details.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The Requester is authenticated and owns the request within their department scope. P2. The request is Submitted or UnderReview and no linked transfer has been dispatched.

Trigger: The Requester chooses to revise the pending request.

Main flow

1. The system verifies revision permission and retrieves the current request version and allocation information.
2. The system confirms that the request is Submitted and has no active reservation before opening its editable fields.
3. The Requester changes the permitted fields, records a revision reason and submits.
4. The system validates the revised fields and rechecks that the request remains Submitted, unreserved and unchanged by another operation.
5. The system saves the new revision in Submitted state and preserves the previous request details and revision reason in its history.
6. The system confirms the revision; subsequent allocation must use the revised requirements.

Alternative flows

A1 at main step 2. The request is UnderReview with an active reservation. The Requester confirms returning the request for revision. The system atomically releases the reservation, restores Reserved units to Available and returns the request to Submitted without changing custody. The system records the return and notifies the reviewer. Return to main step 3 after successful release.

Exception flows

E1 at main step 4. A concurrent operation reserved or approved the request, or the revised quantity is invalid. The system rejects the revision and displays the current state or validation error. It does not overwrite the concurrent operation or retain an approval for changed details. End without saving the revision; a permitted fresh revision may be started.

Success postconditions: The revised request is Submitted and has no active reservation.

Unsuccessful postconditions: No changed requirement is committed against an active allocation; a completed preliminary release remains recorded.

Applicable business rules: BR-01, BR-03, BR-06, BR-08, BR-14.

Functional requirements: P3-FR-11.1, P3-FR-11.2, P3-FR-11.3 (listed in Section 4).

Acceptance check: Revise an UnderReview two-unit request: both reservations are released before editing, custody is unchanged and the new version is Submitted. A request concurrently approved during editing cannot be overwritten.

UC-12 Cancel a resource request

Priority: Essential, later increment. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: End an owned request before dispatch while consistently releasing its physical allocations and retaining its decision history.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The Requester is authenticated and owns the request within their permitted department scope. P2. The request is Draft, Submitted, UnderReview or Approved when cancellation is initiated.

Trigger: The Requester selects cancellation of the request.

Main flow

1. The system checks cancellation permission and retrieves the request with its reservations and linked transfers.
2. The system shows the affected asset IDs and any planned transfer, explaining that cancellation cannot erase completed dispatch or receipt.
3. The Requester enters a cancellation reason and confirms the action.
4. The system rechecks the current request and transfer states and confirms that none of the linked assets has been dispatched.
5. The system atomically sets the request to Cancelled, releases Active reservations, restores affected Reserved units to Available and cancels linked Draft or Scheduled transfers.
6. The system records the cancellation and notifies the relevant Custodian and reviewer, preserving earlier approvals, custody and listing flags.

Alternative flows

A1 at main step 2. The request has no reservation or linked transfer. The system displays that cancellation will affect only the request and its review status. Return to main step 3; step 5 has no asset allocations to release.

Exception flows

E1 at main step 4. A linked transfer has already been Dispatched or Received, including dispatch occurring during confirmation. The system rejects cancellation without releasing allocations or reversing custody. It directs the Requester to the applicable receipt, dispute or loan-return workflow, preserving the dispatched transfer and its history. End unsuccessfully without cancelling the request.

Success postconditions: The request is Cancelled and no pre-dispatch allocation remains active.

Unsuccessful postconditions: The cancellation attempt does not erase dispatch evidence, receipt evidence or authoritative custody.

Applicable business rules: BR-01, BR-05, BR-07, BR-08, BR-14.

Functional requirements: P3-FR-12.1, P3-FR-12.2, P3-FR-12.3 (listed in Section 4).

Acceptance check: Cancel an Approved request with two Reserved units and a Scheduled transfer: both units become Available, the transfer becomes Cancelled and custody is unchanged. The same operation after dispatch is rejected without returning stock to availability.

UC-13 Reserve eligible assets

Priority: Essential, Prototype 1. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Allocate the requested quantity of distinct eligible units for review without granting approval.

Primary actor: Custodian. Supporting actors: Requester.

Preconditions: P1. The Custodian is authenticated and permitted to review the Submitted request. P2. The request exists in Submitted state and has no active reservation.

Trigger: The Custodian begins allocating assets to the submitted need.

Main flow

1. The system retrieves the current request and verifies the Custodian's access to its review information.
2. The system refreshes the request's quantity, mandatory specifications, required date and permitted candidate assets.
3. The Custodian selects exactly the required number of distinct physical Asset IDs and confirms the proposed allocation.
4. Within one transaction, the system rechecks the request version, full quantity and each unit's permission, Available state, listed flag, Serviceable condition and mandatory specifications.
5. The system commits the complete Active reservation, every selected unit's Reserved state and the request's UnderReview state atomically.
6. The system records the allocation, shows its asset IDs and makes the request available for independent approval. Custody and owning departments remain unchanged.

Alternative flows

A1 at main step 3. Fewer eligible units are available than requested. The Custodian leaves the request unallocated and informs the Requester. Only the Requester may revise the quantity through UC-11. Return to main step 2 after revision, or end Submitted and unreserved.

Exception flows

E1 at main step 4. Quantity is incomplete, an ID is repeated, or concurrent changes invalidate the request or a selected unit. The system rejects the entire allocation and reports the conflict. No subset is reserved. Return to main step 2 with refreshed information.

Success postconditions: The request is UnderReview with one complete reservation covering its quantity in distinct Reserved asset IDs.

Unsuccessful postconditions: No partial allocation, approval or custody change is committed.

Applicable business rules: BR-01, BR-02, BR-03, BR-04, BR-05, BR-14.

Functional requirements: P3-FR-13.1, P3-FR-13.2, P3-FR-13.3 (listed in Section 4).

Acceptance check: Two competing requests for the same final monitor cannot both reserve it. For quantity two, selecting the same ID twice or losing one candidate to a concurrent reservation creates no partial allocation.

UC-14 Review a resource request

Priority: Essential, Prototype 1. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Make an accountable decision on a reserved resource request without approving the approver's own request.

Primary actor: Approver. Supporting actors: None.

Preconditions: P1. The approver is signed in. P2. The request exists in UnderReview and is within the approver's assigned department scope.

Trigger: The approver opens a request awaiting a decision.

Main flow

1. The approver selects the request from the review queue.
2. The system checks current approval authority and verifies that the approver is not the requester.
3. The system presents the purpose, quantity, mandatory specifications, required date, transfer type and reserved physical asset IDs.
4. The approver selects approval and records the decision rationale.
5. The system verifies the full set is Reserved by this request, with Serviceable condition, listing, permissions, mandatory specifications and unchanged request details still valid.
6. The system records the approval, approver, timestamp and approved version atomically and sets the request to Approved.
7. The system notifies the requester and custodian and displays the decision record.

Alternative flows

A1 at main step 4. The approver rejects the request. The approver supplies a rejection reason. The system sets Rejected, releases all active reservations and returns eligible reserved assets to Available. End with rejection recorded and affected users notified.

A2 at main step 4. The approver requests a correction. The approver identifies the information requiring revision. The system returns the request to Submitted and releases its reservations. End awaiting requester revision and subsequent reservation and review.

Exception flows

E1 at main step 2. The approver is the requester or lacks current authority. The system denies the decision and records the denied attempt. End without approving or changing the request.

E2 at main step 5. A reserved unit is no longer eligible. The system invalidates the affected reservation set and returns the request to Submitted for reassignment. End without approval; no partial allocation remains.

Success postconditions: One version-specific approval is recorded; all required units remain reserved.

Unsuccessful postconditions: No unauthorized approval exists; the reason for rejection or failed validation is retained.

Applicable business rules: BR-01, BR-02, BR-05, BR-06, BR-08, BR-14.

Functional requirements: P3-FR-14.1, P3-FR-14.2, P3-FR-14.3 (listed in Section 4).

Acceptance check: For a synthetic two-unit request, an independent authorized approver can approve two eligible reserved IDs. Self-approval fails. Rejection releases both reservations and retains its reason.

UC-15 Schedule an asset transfer

Priority: Essential, later increment. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Arrange the authorized movement of the complete reserved asset set without changing custody prematurely.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The custodian is signed in and manages the source assets. P2. An Approved request exists and has not been dispatched.

Trigger: The custodian chooses to schedule the approved request.

Main flow

1. The custodian opens the approved request and its reserved asset list.
2. The system checks source-department authority and displays the approved destination, quantity and loan or permanent-transfer type.
3. The custodian enters the planned handover date, collection point, receiving account and, for a loan, the agreed return date.
4. The system validates the dates, complete active reservation, approved version and absence of another active transfer for each ID.
5. The system displays a Draft transfer summary showing every physical ID and the unchanged current custody.
6. The custodian confirms the schedule.
7. The system saves the transfer as Scheduled and notifies the requester; the assets remain Reserved.

Alternative flows

A1 at main step 1. A Scheduled transfer already exists and has not been dispatched. The system opens the existing transfer rather than creating a duplicate. The custodian requests a revised logistical schedule; the system retains the previous schedule in history. Return to main step 3 for validation and confirmation.

A2 at main step 4. The proposed change alters approved quantity, destination department, specifications or transfer type. The system prevents scheduling against the old approval. On confirmed revision, it cancels the pre-dispatch schedule, releases reservations and returns the request to Submitted. End awaiting revised request approval.

Exception flows

E1 at main step 4. The required date has passed or the reservation is incomplete. The system identifies the failed check and refuses the schedule. The custodian initiates date revalidation or reassignment through the request workflow. End without a dispatchable transfer.

Success postconditions: One Scheduled transfer references the complete approved reservation; ownership and custody are unchanged.

Unsuccessful postconditions: No invalid schedule becomes dispatchable; previously recorded history remains available.

Applicable business rules: BR-01, BR-05, BR-06, BR-07, BR-08, BR-15.

Functional requirements: P3-FR-15.1, P3-FR-15.2, P3-FR-15.3 (listed in Section 4).

Acceptance check: Scheduling two reserved synthetic units produces one Scheduled transfer without changing custody. A changed destination department requires resubmission; a repeated scheduling attempt cannot create a second active transfer.

UC-16 Release reserved assets

Priority: Essential, Prototype 1. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Record dispatch only when every physical unit and the current approval permit the planned movement.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The custodian is signed in and assigned to the source department. P2. The transfer is Scheduled and has not previously been dispatched.

Trigger: The custodian prepares the scheduled asset set for handover.

Main flow

1. The custodian opens the Scheduled transfer.
2. The system verifies release authority and displays the required physical IDs and receiving details.
3. The custodian checks each unit against its ID and records handover evidence, dispatch time and recipient details.
4. The system rechecks approval, full reservation, serviceability, mandatory specifications and the required date against current records.
5. The system displays the complete dispatch record and warns that cancellation will no longer be available after dispatch.
6. The custodian confirms the handover.
7. The system atomically sets the transfer to Dispatched, consumes its reservations and sets all units to InTransit while retaining authoritative custody.
8. The system records the dispatch event and notifies the requester to confirm receipt.

Alternative flows

A1 at main step 3. One physical ID does not match the reserved set. The system identifies the mismatch. The custodian locates the correct unit rather than substituting an unapproved asset. Return to main step 3; no partial release is recorded.

Exception flows

E1 at main step 4. A unit is faulted or another material approval condition has changed. The system blocks the entire dispatch and invalidates the approval. It releases reservations, returns the request to Submitted and cancels the pre-dispatch transfer; faulted units remain unavailable. End awaiting reassignment and independent approval.

E2 at main step 4. The required date has passed. The system blocks dispatch pending recorded date revalidation. End without changing custody or consuming reservations.

Success postconditions: All specified units are InTransit under one recorded dispatch; the request is not yet Fulfilled.

Unsuccessful postconditions: No partial dispatch is committed; unauthorized or ineligible release is blocked.

Applicable business rules: BR-01, BR-05, BR-06, BR-07, BR-08, BR-09, BR-14.

Functional requirements: P3-FR-16.1, P3-FR-16.2, P3-FR-16.3 (listed in Section 4).

Acceptance check: Dispatching two eligible synthetic units sets both InTransit and consumes both reservations. Faulting either unit before confirmation blocks the complete dispatch; neither unit receives a partial dispatch record.

UC-17 Confirm asset receipt

Priority: Essential, Prototype 1. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Establish evidenced custody of the complete delivery, including controlled acceptance of damaged units.

Primary actor: Requester. Supporting actors: Custodian, Maintenance Officer.

Preconditions: P1. The requester is signed in as the recorded receiver. P2. A Dispatched or Disputed transfer exists with its original dispatch evidence.

Trigger: The requester records the result of receiving the delivery.

Main flow

1. The requester opens the transfer and reviews the dispatched physical IDs.
2. The system verifies receiving authority and displays incidents and previous custody.
3. The requester checks every unit and enters receipt time, location, condition and supporting evidence.
4. The system validates the complete ID set, serviceable condition and absence of unresolved incidents.
5. The requester confirms acceptance of the complete delivery.
6. The system atomically sets Received, updates custody and location, sets the units InUse and marks the request Fulfilled; a permanent transfer also changes owning department, whereas a loan does not.
7. The system preserves dispatch and receipt events and notifies the source custodian.

Alternative flows

A1 at main step 3. Units are missing, different or damaged. The requester records the discrepancy and evidence. The system sets Disputed and preserves InTransit status and previous custody for the transfer set. End pending an evidenced resolution; the request remains unfulfilled.

A2 at main step 4. A recorded dispute has been resolved. The receiver and dispatching custodian confirm the resolution and complete ID set. For damage, a Maintenance Officer records an inspection and disposition permitting receipt into UnderMaintenance. The requester confirms receipt; the system applies step 6, but damaged units remain unlisted and UnderMaintenance rather than InUse. Return to main step 7 with the original dispute and timestamps retained.

Exception flows

E1 at main step 4. An incident remains unresolved or resolution evidence is missing. The system blocks receipt confirmation and identifies the outstanding evidence. End with InTransit status and previous custody unchanged.

Success postconditions: All IDs have confirmed receipt; ownership follows the transfer type and unsafe units remain unavailable.

Unsuccessful postconditions: Unresolved delivery discrepancies never silently change custody or fulfil the request.

Applicable business rules: BR-01, BR-07, BR-09, BR-14, BR-15.

Functional requirements: P3-FR-17.1, P3-FR-17.2, P3-FR-17.3 (listed in Section 4).

Acceptance check: A two-unit delivery missing one ID stays Disputed and unfulfilled. After evidenced resolution, both IDs receive custody updates; an accepted damaged unit is UnderMaintenance and unlisted, not serviceable stock.

UC-18 Return loaned assets

Priority: Essential, later increment. Parent group: FR-04. Proposed verification owner: Saeed Alkaabi.

Goal: Restore original custody while preserving ownership and loan history.

Primary actor: Requester. Supporting actors: Custodian, Maintenance Officer.

Preconditions: P1. The requester is signed in and holds loan custody. P2. The original loan transfer is Received; selected units remain outstanding.

Trigger: The requester initiates a return against the original loan.

Main flow

1. The requester selects the outstanding loan and the complete set to be returned.
2. The system verifies current custody, original owning department and absence of a conflicting active transfer.
3. The requester records a proposed handover date, current condition and location.
4. The original custodian confirms the receiving arrangement.
5. The system creates a Scheduled return transfer linked to the original loan; it creates no new resource request.
6. The requester confirms dispatch evidence; the system sets Dispatched and InTransit, retaining borrower custody.
7. The original custodian checks all physical IDs, records inspection evidence and confirms receipt of the serviceable set.
8. The system sets Received, restores original custody and location, sets Available but unlisted, and records the return without changing ownership.

Alternative flows

A1 at main step 1. The linked return is already Disputed. The system opens that return and retains its dispatch, dispute and custody history. Both parties record resolution evidence against the original expected IDs; the system creates no new return. Return to main step 7 for ID verification and receipt inspection.

A2 at main step 7. A returned unit is damaged or already carries a fault flag. The system records a dispute pending inspection and keeps previous custody. After a Maintenance Officer's disposition and both parties' evidenced resolution, the custodian confirms receipt into UnderMaintenance. Return to main step 8, replacing Available with UnderMaintenance for affected units; all remain unlisted.

Exception flows

E1 at main step 2. The unit belongs to another loan, was already returned or has a conflicting transfer. The system refuses the return and identifies the incompatible record. End without a duplicate return or custody change.

E2 at main step 7. An expected unit is missing. The system records Disputed and preserves InTransit status and borrower custody until the discrepancy is resolved. End without confirming the complete return.

Success postconditions: The return is linked to its loan; receipt restores original custody without automatic publication.

Unsuccessful postconditions: The loan history and owning department remain unchanged; unresolved returns do not erase borrower custody.

Applicable business rules: BR-01, BR-07, BR-09, BR-14, BR-15.

Functional requirements: P3-FR-18.1, P3-FR-18.2, P3-FR-18.3 (listed in Section 4).

Acceptance check: Returning a synthetic loan restores original custody only on receipt. Ownership never changes. A damaged return enters UnderMaintenance; attempting to return the same ID again creates no second return.

UC-19 Record an asset inspection

Priority: Essential, Prototype 1. Parent group: FR-05. Proposed verification owner: Ali Alawadhi.

Goal: Record inspection evidence and safety consequences without silently resolving transfers or maintenance jobs.

Primary actor: Maintenance Officer. Supporting actors: None.

Relationships: This use case is included by UC-22 Complete asset maintenance.

Preconditions: P1. The Maintenance Officer is signed in and has access to the asset. P2. A non-retired asset record exists.

Trigger: The officer records an inspection performed on a synthetic asset scenario.

Main flow

1. The officer opens the asset and starts a new inspection record.
2. The system displays its current state, custody, condition and linked maintenance or transfer records.
3. The officer records the date, checks, findings, Serviceable condition and passing evidence.
4. The system validates the asset ID, required inspection fields and evidence references.
5. The officer reviews and confirms the inspection.
6. The system appends the inspection and updates condition to Serviceable without automatically completing maintenance, resolving a dispute or publishing the asset.
7. The system records the inspector and timestamp and makes the result available to permitted workflow participants.

Alternative flows

A1 at main step 3. Inspection fails or records NeedsInspection or Unserviceable. After validation, the officer confirms failed checks and evidence; the system saves the reported condition and unlists the asset. Available, Reserved or InUse units enter UnderMaintenance without changing custody. For Reserved, it releases the whole reservation, restores other released serviceable units to Available, cancels the pre-dispatch schedule, invalidates approval and returns the request to Submitted. For InTransit, it preserves custody and state and records a blocking incident. Return to main step 7 after recording the safety action.

A2 at main step 1. An earlier inspection contains an error. The officer identifies the earlier record and a correction reason. The system links the new inspection as a correction rather than replacing historical evidence. Return to main step 2.

Exception flows

E1 at main step 4. The result lacks required evidence or contains an invalid asset reference. The system marks the missing or invalid fields and saves no inspection outcome. Return to main step 3.

Success postconditions: The inspection is attributable and immutable; the appropriate safety restriction is recorded.

Unsuccessful postconditions: Invalid inspection inputs do not certify serviceability or alter custody.

Applicable business rules: BR-01, BR-09, BR-14.

Functional requirements: P3-FR-19.1, P3-FR-19.2, P3-FR-19.3 (listed in Section 4).

Acceptance check: Failing inspection on a reserved synthetic ID returns its request to Submitted and prevents dispatch. Failing an InTransit ID preserves custody and opens an incident. A later passing inspection alone does not publish it.

UC-20 Report an asset fault

Priority: Essential, Prototype 1. Parent group: FR-05. Proposed verification owner: Ali Alawadhi.

Goal: Prevent unsafe allocation after a fault without rewriting custody or movement history.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The requester is signed in and has permitted access to a non-retired asset record.

Trigger: The requester observes a fault and selects the asset's fault-report action.

Main flow

1. The requester identifies the asset by its physical ID.
2. The system verifies access and displays its recorded condition and current workflow state.
3. The requester enters observed symptoms, observation time, location and available supporting evidence.
4. The system validates the asset reference and required observations without asking the requester to certify a repair diagnosis.
5. The requester confirms the report.
6. For the normal Available-asset path, the system records NeedsInspection, removes the listing and places the asset UnderMaintenance.
7. The system appends the fault event, preserves custody and notifies the responsible custodian and Maintenance Officer.

Alternative flows

A1 at main step 6. The asset is Reserved. The system unlists the unit and sets UnderMaintenance. It releases the complete reservation, invalidates approval, returns the request to Submitted and cancels any pre-dispatch schedule. Return to main step 7, notifying the requester about reassignment.

A2 at main step 6. The asset is InUse, UnderMaintenance or Draft. The system retains custody and records NeedsInspection. InUse becomes UnderMaintenance; existing UnderMaintenance or Draft remains unchanged. Publication stays blocked. Return to main step 7.

A3 at main step 6. The asset is InTransit. The system records a blocking receipt incident and NeedsInspection. It preserves the transfer, InTransit status and previous authoritative custody. Return to main step 7; resolution follows UC-17 or UC-18.

Exception flows

E1 at main step 4. The description is empty. The system refuses submission and identifies the missing observation. Return to main step 3 without saving a fault record.

Success postconditions: The fault is attributable; the asset cannot remain eligible for normal allocation or unqualified receipt.

Unsuccessful postconditions: An invalid report changes no authoritative record; existing faults and history remain intact.

Applicable business rules: BR-01, BR-05, BR-07, BR-09, BR-14.

Functional requirements: P3-FR-20.1, P3-FR-20.2, P3-FR-20.3 (listed in Section 4).

Acceptance check: Faulting one of two reserved synthetic units releases the whole reservation and returns the request to Submitted. Faulting a dispatched unit instead blocks receipt without rewriting custody or cancelling its history.

UC-21 Plan asset maintenance

Priority: Essential, later increment. Parent group: FR-05. Proposed verification owner: Ali Alawadhi.

Goal: Schedule and start authorized maintenance while keeping affected assets unavailable.

Primary actor: Maintenance Officer. Supporting actors: None.

Preconditions: P1. The Maintenance Officer is signed in with asset access. P2. A fault, inspection finding or approved repair or refurbishment action exists; the asset is UnderMaintenance or Available and Serviceable.

Trigger: The officer plans or starts work for the recorded maintenance need.

Main flow

1. The officer selects the supporting fault, inspection or lifecycle authorization.
2. For the UnderMaintenance asset, the system displays existing jobs and prepares an Open record.
3. The officer enters the work, responsible Maintenance Officer, dates and any estimated cost with its basis.
4. The system validates dates, assigned role, nonnegative costs and absence of conflicting active work.
5. The officer confirms the work plan.
6. The system saves the job as Scheduled, links its supporting evidence and keeps the asset UnderMaintenance and unlisted.
7. The system records the planner and timestamp and notifies the assigned officer and custodian.

Alternative flows

A1 at main step 2. The assigned officer starts an existing Scheduled job. The system rechecks assignment, job version and UnderMaintenance status. The officer records the actual start time; the system changes that job to InProgress without duplication. Return to main step 7.

A2 at main step 2. An Open or Scheduled job already addresses the same need. The officer opens that job instead of creating a duplicate. The system retains its prior schedule and the reason for revision. Return to main step 3.

A3 at main step 2. An Available, Serviceable asset has approved repair or refurbishment authority. The system verifies current owning-department custody and no outstanding loan, reservation, active transfer or unresolved incident. It records the authorized maintenance hold, unlists the asset, sets UnderMaintenance and prepares the Open job. Return to main step 3.

A4 at main step 5. The planned job is no longer required. The officer records a cancellation reason. The system marks the job Cancelled without clearing the fault or making the asset available. End with cancellation history retained.

Exception flows

E1 at main step 2. The requested maintenance hold or start fails an authority, custody or conflict check. The system refuses the transition without creating a duplicate job or changing asset state. End awaiting resolution of the blocking condition.

E2 at main step 4. The assigned account lacks the role, dates are inconsistent or a supplied cost is negative. The system identifies each failed validation and does not schedule the job. Return to main step 3.

Success postconditions: One attributed maintenance plan exists with an owner, dates and evidence; the asset stays unavailable.

Unsuccessful postconditions: No invalid or conflicting job is scheduled; cancellation does not certify serviceability.

Applicable business rules: BR-01, BR-03, BR-09, BR-14.

Functional requirements: P3-FR-21.1, P3-FR-21.2, P3-FR-21.3 (listed in Section 4).

Acceptance check: Approved refurbishment of an unallocated synthetic unit sets UnderMaintenance and unlists it. Starting its Scheduled job sets InProgress once. An outstanding loan blocks the new hold; cancellation never restores availability.

UC-22 Complete asset maintenance

Priority: Essential, Prototype 1. Parent group: FR-05. Proposed verification owner: Ali Alawadhi.

Goal: Close maintenance and restore service only with inspection evidence.

Primary actor: Maintenance Officer. Supporting actors: None.

Relationships: This use case includes UC-19 Record an asset inspection.

Preconditions: P1. The assigned Maintenance Officer is signed in. P2. The maintenance job is InProgress and its asset is UnderMaintenance.

Trigger: The officer records that the maintenance activity has finished.

Main flow

1. The officer opens the job and reviews the fault and planned work.
2. The officer records work performed, completion time, actual nonnegative costs and a linked passing inspection with evidence.
3. The system validates the job version, required completion fields, inspection result and current custody.
4. The system displays the completion summary and the proposed post-maintenance state.
5. The officer confirms completion.
6. For the normal owner-held asset, the system marks the job Completed, sets condition Serviceable and restores Available while leaving the asset unlisted.
7. The system appends completion, preserves earlier evidence and notifies the custodian.

Alternative flows

A1 at main step 2. The completed repair attempt failed its inspection. The officer records the failed result, work and actual costs. After validation and confirmation, the system marks the job Completed with an unsuccessful outcome while retaining UnderMaintenance and the failed condition. Return to main step 7; further work or a lifecycle proposal is required before reuse.

A2 at main step 6. The repaired unit remains in a borrower's confirmed custody. The system marks the job Completed, sets condition Serviceable and restores InUse rather than Available, leaving the asset unlisted. It retains the original owning department and outstanding loan. Return to main step 7.

Exception flows

E1 at main step 3. A passing outcome lacks an inspection or recorded cost is negative. The system refuses completion and identifies the missing or invalid value. Return to main step 2; no service restoration occurs.

E2 at main step 6. The job has already been completed. The system displays the existing completion rather than duplicating costs or state changes. End with one completion record.

Success postconditions: The completed job has evidence and costs; its result determines availability without changing custody or publishing.

Unsuccessful postconditions: Missing or failed inspection cannot silently return an unsafe asset to service.

Applicable business rules: BR-01, BR-03, BR-09, BR-14, BR-15.

Functional requirements: P3-FR-22.1, P3-FR-22.2, P3-FR-22.3 (listed in Section 4).

Acceptance check: A repaired owner-held synthetic asset becomes Available but unlisted. The same passing repair on a borrowed unit restores InUse. A failed inspection closes the attempt without clearing UnderMaintenance.

UC-23 Propose a lifecycle action

Priority: Essential, later increment. Parent group: FR-05. Proposed verification owner: Humaid Al Ali.

Goal: Submit an evidenced lifecycle proposal for independent review.

Primary actor: Custodian. Supporting actors: None.

Relationships: This use case is extended by UC-33 Recommend a sustainable action.

Preconditions: P1. The custodian is signed in and manages the asset. P2. The non-retired asset has an existing record and accessible lifecycle history.

Trigger: The custodian identifies a lifecycle action for consideration.

Main flow

1. The custodian opens the asset and its condition, maintenance and transfer history.
2. The system checks proposal authority and displays existing lifecycle proposals and movement restrictions.
3. The custodian selects one action type and records a rationale, supporting evidence and any known cost or mass inputs with units and sources.
4. The system validates the action type, required rationale, evidence references and nonnegative supplied numerical inputs.
5. The system presents the proposed action and states that submission does not authorize execution or change asset ownership.
6. The custodian confirms submission.
7. The system creates a Proposed lifecycle action linked to the current asset version, records its author and timestamp, and notifies the authorized approver.

Alternative flows

A1 at main step 3. An AI recommendation informed the proposal. The custodian reviews its rationale and limitations and explicitly chooses the proposed action. The system links the recommendation and its version trace; the custodian records reasons for any override. Return to main step 4; AI advice confers no approval authority.

A2 at main step 3. A cost, mass or benefit input is unknown. The custodian marks the input unavailable and states the evidence needed. The system keeps it missing rather than filling zero or inventing a benefit. Return to main step 4 with the limitation visible to the approver.

Exception flows

E1 at main step 4. The rationale is absent or a supplied cost or mass is negative. The system identifies the invalid fields and refuses submission. Return to main step 3.

Success postconditions: An attributable Proposed action exists for independent review; asset state, custody and ownership are unchanged.

Unsuccessful postconditions: Invalid or unconfirmed advice cannot become an authorized action or retire the asset.

Applicable business rules: BR-01, BR-03, BR-10, BR-11, BR-12, BR-13, BR-14.

Functional requirements: P3-FR-23.1, P3-FR-23.2, P3-FR-23.3 (listed in Section 4).

Acceptance check: A synthetic recycling proposal with rationale becomes Proposed without retiring the asset. Missing mass remains unavailable. An AI suggestion alone cannot submit or approve the proposal, and negative recorded mass is rejected.

UC-24 Authorize a lifecycle action

Priority: Essential, later increment. Parent group: FR-05. Proposed verification owner: Humaid Al Ali.

Goal: Independently accept or reject a specific lifecycle proposal while keeping execution a separate evidenced action.

Primary actor: Approver. Supporting actors: None.

Preconditions: P1. The approver is signed in. P2. A Proposed lifecycle action exists within the approver's assigned review scope.

Trigger: The approver opens the lifecycle proposal for a decision.

Main flow

1. The approver selects the proposal and its linked asset.
2. The system verifies current business approval authority and that the approver did not propose the action.
3. The system displays the action, rationale, supporting inspections, known costs, missing inputs and any AI advice separately from evidence.
4. The approver selects approval and records the decision rationale.
5. The system verifies the reviewed proposal version and identifies the permitted next workflow for the selected action.
6. The system sets the lifecycle action to Approved and records the approver, decision time and authorized version without changing the asset state.
7. The system notifies the custodian: reuse follows the request and transfer workflow, repair or refurbishment follows maintenance, and donation, recycling or disposal requires retirement execution evidence.

Alternative flows

A1 at main step 4. The approver rejects the proposal. The approver enters a rejection reason. The system sets Rejected and notifies the custodian without altering the asset. End with an attributable rejection.

A2 at main step 4. Evidence is insufficient for a decision. The approver identifies the missing evidence and requests clarification. The system retains Proposed status and records the clarification request. End awaiting additional evidence; there is no execution authority.

Exception flows

E1 at main step 2. The approver is the proposer or has only Administrator privileges. The system denies approval and records the attempted action. End with the proposal still unapproved.

E2 at main step 5. Material proposal details changed after review. The system blocks the decision against the stale version and presents the revised evidence. Return to main step 3 for a fresh review.

Success postconditions: One independent version-specific authorization exists; execution and its evidence remain outstanding.

Unsuccessful postconditions: No self-approval, administrative bypass or unsupported retirement occurs.

Applicable business rules: BR-01, BR-02, BR-10, BR-11, BR-14.

Functional requirements: P3-FR-24.1, P3-FR-24.2, P3-FR-24.3 (listed in Section 4).

Acceptance check: An independent Approver can authorize a synthetic disposal proposal but the asset does not become Retired. Self-approval and an Administrator without Approver authority fail. A repair approval routes to maintenance instead of retirement.

UC-25 Record asset retirement

Priority: Essential, later increment. Parent group: FR-05. Proposed verification owner: Humaid Al Ali.

Goal: Record evidenced execution of authorized donation, recycling or disposal.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The custodian is signed in and manages the asset. P2. An Approved lifecycle proposal exists for the non-retired asset.

Trigger: The custodian records execution of the approved retirement action in the synthetic prototype.

Main flow

1. The custodian opens the approved donation, recycling or disposal action and checks the physical asset ID.
2. The system verifies execution authority, independent approval and the unchanged authorized proposal version.
3. The custodian records the execution date, simulated recipient or operator, method and supporting synthetic evidence.
4. The system validates evidence, confirms custody in the current owning department, and checks for outstanding loans, reservations, transfers and unresolved transit incidents.
5. The system presents the proposed retirement and retained historical records for confirmation.
6. The custodian confirms that the recorded scenario matches the authorized action.
7. The system atomically sets the lifecycle action to Executed and the asset to Retired and unlisted, then records the event and notifies permitted stakeholders.

Alternative flows

A1 at main step 1. The selected authorization is for reuse, repair or refurbishment. The system does not offer retirement execution. It directs the custodian to the transfer or maintenance workflow linked to that action. End without retiring the asset.

Exception flows

E1 at main step 4. Evidence is missing, the unit is borrowed or an active movement remains unresolved. The system identifies the blocking condition and refuses retirement. The custodian resolves the underlying receipt, return or evidence issue before retrying. End with the asset state and custody unchanged.

E2 at main step 7. Execution was already recorded. The system displays the existing execution record without adding another retirement or benefit event. End with one authoritative retirement record.

Success postconditions: The physical ID is Retired and unlisted with linked authorization and execution evidence; previous history remains accessible.

Unsuccessful postconditions: No asset is retired without evidence or while custody is unresolved; no real external ownership transfer is asserted.

Applicable business rules: BR-01, BR-02, BR-07, BR-09, BR-10, BR-14, BR-16.

Functional requirements: P3-FR-25.1, P3-FR-25.2, P3-FR-25.3 (listed in Section 4).

Acceptance check: An approved synthetic recycling action with execution evidence retires and unlists its ID. An outstanding loan or missing evidence blocks retirement. Repeating confirmation neither creates a second event nor duplicates benefit eligibility.

UC-26 Generate an operational report

Priority: Essential, Prototype 1. Parent group: FR-06. Proposed verification owner: Humaid Al Ali.

Goal: Produce a reproducible view of permitted assets and workflow activity without changing operational records.

Primary actor: Custodian. Supporting actors: None.

Preconditions: P1. The custodian is signed in with reporting access for an assigned department.

Trigger: The custodian requests an operational report.

Main flow

1. The custodian selects a report type, department scope, period and applicable status filters.
2. The system validates filter values and identifies the records the custodian may access.
3. The system establishes a report cutoff and retrieves permitted records for the selected criteria.
4. The system calculates counts from distinct record identifiers and attaches the selected filters to the result.
5. The system displays the report with its generation time, cutoff and links to permitted supporting records.
6. The custodian reviews the result and requests an export.
7. The system rechecks export access and produces the same scoped report, excluding protected fields.

Alternative flows

A1 at main step 4. No records match the validated filters. The system displays zero matching records and the applied filters without inventing report rows. Return to main step 5.

Exception flows

E1 at main step 3. The reporting query cannot complete consistently. The system labels the report unavailable and does not present a partial count as a completed total. The system offers a retry using the same filters. End without producing a completed report.

Success postconditions: A dated, scoped report and requested export are available. Operational records remain unchanged.

Unsuccessful postconditions: No misleading completed report is issued; source records remain unchanged.

Applicable business rules: BR-01, BR-14.

Functional requirements: P3-FR-26.1, P3-FR-26.2, P3-FR-26.3 (listed in Section 4).

Acceptance check: A report for a synthetic department reconciles with its distinct permitted records. A record from another restricted department does not appear in its rows, totals or export.

UC-27 Record benefit calculation inputs

Priority: Essential, later increment. Parent group: FR-06. Proposed verification owner: Humaid Al Ali.

Goal: Record traceable financial evidence for completed qualifying activity so later reports can calculate benefits without unsupported assumptions.

Primary actor: Finance Officer. Supporting actors: None.

Preconditions: P1. The finance officer is signed in with access to benefit records for the relevant department.

Trigger: The finance officer opens a completed event for benefit assessment.

Main flow

1. The finance officer selects the completed event and identifies its physical asset IDs.
2. The system checks event completion, department access and existing benefit claims for the same qualifying activity.
3. The finance officer enters the avoided-purchase baseline, repair and transfer costs in AED, evidence references and assessment date.
4. The system validates the amounts and supporting references and displays the proposed calculation inputs.
5. The finance officer confirms whether the evidence supports an actual avoided purchase and supplies the assessment reason.
6. The system saves a versioned benefit-input record linked to the event and records who confirmed the assessment.
7. The system displays the recorded inputs and their availability for subsequent financial reporting.

Alternative flows

A1 at main step 4. A baseline or required cost is unknown. The system records the missing field as unavailable, together with the reason. The finance officer saves an incomplete assessment without confirming an avoided-purchase value. End with financial benefit unavailable for this event.

Exception flows

E1 at main step 6. Another assessment has already claimed the same qualifying event. The system rejects duplicate creation and displays the existing permitted assessment. A correction must create a traceable revision rather than a second claim. End without adding duplicate benefit.

Success postconditions: Financial inputs have evidence references and a recorded assessment status.

Unsuccessful postconditions: No duplicate claim or unsupported completed calculation is created.

Applicable business rules: BR-01, BR-03, BR-12, BR-14.

Functional requirements: P3-FR-27.1, P3-FR-27.2, P3-FR-27.3, P3-FR-27.4 (listed in Section 4).

Acceptance check: Validated inputs of AED 800 avoided purchase, AED 150 repair and AED 50 transfer support AED 600 benefit. Other currencies are rejected. A missing baseline produces unavailable benefit, and repeating the event cannot increase the total.

UC-28 Generate a sustainability report

Priority: Essential, Prototype 1. Parent group: FR-06. Proposed verification owner: Humaid Al Ali.

Goal: Report recorded reuse, repair and resource outcomes with transparent financial and environmental calculation evidence.

Primary actor: Sustainability Officer. Supporting actors: None.

Preconditions: P1. The sustainability officer is signed in with access to the selected departments' sustainability records.

Trigger: The officer requests a sustainability report for a selected period.

Main flow

1. The officer chooses department scope, period and required indicators.
2. The system validates the filters and retrieves completed qualifying events with their distinct asset IDs.
3. The officer reviews the associated mass evidence, dated environmental factors, units and financial assessments available for calculation.
4. The system validates provenance, deduplicates qualifying events and prevents repeated physical-mass credit within the same indicator and period.
5. The system calculates supported reuse, repair, waste-diversion, life-extension, savings and estimated carbon indicators.
6. The system displays values with formulas, denominators, evidence coverage and a distinction between recorded measurements and estimates.
7. The officer reviews the supporting records and exports the dated report with its assumptions and limitations.

Alternative flows

A1 at main step 5. An indicator lacks required evidence, a valid denominator or a supported factor. The system marks that indicator unavailable and identifies the missing inputs. Other supported indicators remain visible, with incomplete coverage disclosed. Return to main step 6.

Exception flows

E1 at main step 4. An environmental factor has incompatible units or no identifiable source and date. The system excludes the factor, explains the failure and marks the affected indicator unavailable. It substitutes no guessed conversion or carbon result. Return to main step 5 for the remaining supported indicators.

Success postconditions: The report distinguishes supported results, estimates and unavailable indicators.

Unsuccessful postconditions: Unsupported environmental or financial values are not reported as established outcomes.

Applicable business rules: BR-01, BR-03, BR-12, BR-13.

Functional requirements: P3-FR-28.1, P3-FR-28.2, P3-FR-28.3, P3-FR-28.4 (listed in Section 4).

Acceptance check: A completed synthetic reuse event appears once. Removing its supported carbon factor makes its carbon estimate unavailable while preserving its reuse count; an unconfirmed transfer contributes no completed-transfer benefit.

UC-29 Read status notifications

Priority: Essential, Prototype 1. Parent group: FR-06. Proposed verification owner: Ali Alawadhi.

Goal: Allow each account role to review relevant workflow updates through its own protected inbox.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The requester is signed in; the same inbox behavior applies to other assigned roles.

Trigger: The requester opens the notification inbox.

Main flow

1. The system retrieves notifications addressed to the signed-in account.
2. The system displays each notification's event type, occurrence time, permitted summary and read status.
3. The requester selects a notification about an asset request or related workflow.
4. The system displays its recorded update and rechecks access to the linked operational record.
5. The requester opens the permitted record to inspect its current state.
6. The system displays the current record separately from the notification's historical event and marks the notification read.

Alternative flows

A1 at main step 4. The linked workflow has changed since the notification occurred. The system retains the original event time and identifies that a newer state exists. The notification does not reinstate an earlier approval, reservation or assignment. Return to main step 5.

A2 at main step 1. The account has no notifications. The system displays an empty inbox without implying a workflow error. End with no notification selected.

Exception flows

E1 at main step 4. The account no longer has permission to open the linked record. The system denies the link and withholds currently restricted record details. It explains that the update cannot be opened with current access. End without exposing the restricted record.

Success postconditions: The requester reads an addressed notification and, where permitted, the current linked record. Only notification read status changes.

Unsuccessful postconditions: No unrelated account's notification or restricted operational details are disclosed.

Applicable business rules: BR-01.

Functional requirements: P3-FR-29.1, P3-FR-29.2, P3-FR-29.3 (listed in Section 4).

Acceptance check: A synthetic rejection notification opens the permitted request and remains historically identifiable after a later update. A different account cannot read it, and reading never changes the request's state.

UC-30 Review the audit history

Priority: Essential, later increment. Parent group: FR-06. Proposed verification owner: Ali Alawadhi.

Goal: Trace authorized operational decisions and corrections without allowing the reviewer to rewrite history.

Primary actor: Auditor. Supporting actors: None.

Preconditions: P1. The auditor is signed in with a defined audit-access scope.

Trigger: The auditor requests the history of an asset, request, transfer or other permitted record.

Main flow

1. The auditor supplies a record identifier, event period and optional action filter.
2. The system validates the filters and checks the auditor's permitted record scope.
3. The system retrieves matching audit events in their recorded sequence.
4. The system displays event identity, time, responsible account, action, outcome and associated record versions.
5. The auditor selects an event to compare its permitted before-and-after values and linked approval or correction evidence.
6. The system presents the linked history without editable event controls.
7. The auditor exports the scoped history with its filters and generation time.

Alternative flows

A1 at main step 3. No events match the valid filters. The system displays that no matching events were found and retains the filters for revision. End without inventing a history entry.

A2 at main step 5. The selected event has a subsequent correction. The system displays the original event and the separately attributed correction together. Return to main step 6.

Exception flows

E1 at main step 2. The supplied identifier belongs to a record outside the auditor's scope. The system denies access without displaying the restricted record's event details. Changing filters cannot override the access restriction. End without releasing the requested history.

Success postconditions: The auditor can trace permitted decisions and corrections while original events remain unchanged.

Unsuccessful postconditions: Restricted history remains undisclosed; no audit entry is altered or deleted.

Applicable business rules: BR-01, BR-14.

Functional requirements: P3-FR-30.1, P3-FR-30.2, P3-FR-30.3 (listed in Section 4).

Acceptance check: A synthetic asset correction displays both the original change and its attributed correction. Audit viewing offers no edit or delete operation, and a restricted department's history is denied.

UC-31 Suggest asset classifications

Priority: Essential, Prototype 1. Parent group: AI-01. Proposed verification owner: Humaid Al Ali.

Goal: Suggest classifications for human review without changing authoritative asset data.

Primary actor: Custodian. Supporting actors: None.

Relationships: This use case extends UC-04 Register an asset; includes UC-35 Review an AI recommendation.

Preconditions: P1. The custodian is signed in and may inspect the selected synthetic asset. P2. An authorized classification configuration is available.

Trigger: The custodian requests classification suggestions for an asset draft or permitted record.

Main flow

1. The custodian selects the asset title, description, specifications and approved JPEG/PNG photographs or text-based PDF documents.
2. The system validates access and files, analyzes photographs and extracts readable PDF text; scanned-document OCR is outside the proposed boundary.
3. The system prepares synthetic inputs, treating documents as data rather than instructions.
4. The classification service produces suggested category, subcategory, tags, description, possible material and missing-metadata observations.
5. The system validates suggested values against reference data and identifies uncertainty or unsupported suggestions.
6. The system displays the candidate classification with its explanation and model, prompt or rule version.
7. The custodian opens human review through UC-35; the asset fields remain unchanged until separately confirmed.

Alternative flows

A1 at main step 5. Image interpretation, text extraction or classification is uncertain. The system withholds unsupported labels and requests human review of the uncertain evidence. The custodian may supply clearer supported inputs or missing metadata. End with an advisory clarification request and unchanged asset fields.

Exception flows

E1 at main step 4. The classification service fails or returns an invalid response. The system reports that suggestions are unavailable. Manual classification through the ordinary asset workflow remains available. End without applying a classification.

Success postconditions: A traceable advisory classification is available for human review.

Unsuccessful postconditions: Service errors or uncertainty do not corrupt the asset record.

Applicable business rules: BR-01, BR-11, BR-16.

Functional requirements: P3-FR-31.1, P3-FR-31.2, P3-FR-31.3, P3-FR-31.4 (listed in Section 4).

Acceptance check: Supported synthetic text, photograph and PDF inputs produce reviewable suggestions. Uncertain extraction requests review; unsupported categories are withheld. Service failure permits manual entry, and no suggestion silently changes the asset.

UC-32 Match resources semantically

Priority: Essential, Prototype 1. Parent group: AI-02. Proposed verification owner: Humaid Al Ali.

Goal: Rank meaningful resource matches without bypassing mandatory eligibility.

Primary actor: Requester. Supporting actors: None.

Relationships: This use case extends UC-08 Search the asset catalogue.

Preconditions: P1. The requester is signed in with access to the selected synthetic request or search criteria. P2. An authorized matching configuration is available.

Trigger: The requester asks for semantic matches.

Main flow

1. The requester selects a request and confirms its purpose, mandatory specifications, quantity and required date.
2. The system validates the request inputs and the requester's department scope.
3. The system constructs candidates from permitted, listed, Available and Serviceable physical asset IDs meeting every mandatory specification.
4. The matching service compares the meaning of the request with eligible candidate descriptions.
5. The system ranks suitable results using proposed weights for purpose, technical fit, condition, quantity, location, urgency, required date, costs, waste diversion and carbon estimates where supported inputs exist.
6. The system displays explanatory scores, missing inputs and configuration versions; scores are not proven probabilities.
7. The requester selects candidate IDs for Custodian review through UC-13; this selection does not reserve them.

Alternative flows

A1 at main step 3. Fewer eligible physical IDs exist than the requested quantity. The system explains that full fulfilment is unavailable and does not reserve a partial set. A permitted request revision may reduce quantity before a new matching attempt. End without a reservation.

Exception flows

E1 at main step 4. The semantic service is unavailable. The system labels semantic results unavailable and offers structured search over the same eligible scope. The fallback does not present keyword results as AI rankings. End semantic matching without altering the request or assets.

Success postconditions: The requester receives explained, eligible suggestions; no reservation or approval is created by ranking.

Unsuccessful postconditions: Ineligible or insufficient units do not become an approved allocation.

Applicable business rules: BR-01, BR-03, BR-04, BR-05, BR-11.

Functional requirements: P3-FR-32.1, P3-FR-32.2, P3-FR-32.3 (listed in Section 4).

Acceptance check: A highly similar but Unserviceable asset never appears as eligible. A request for two units cannot be fulfilled using one ID twice, and selecting a result alone does not reserve it.

UC-33 Recommend a sustainable action

Priority: Desirable. Parent group: AI-03. Proposed verification owner: Humaid Al Ali.

Goal: Compare supported lifecycle options while keeping asset decisions and execution under authorized human control.

Primary actor: Custodian. Supporting actors: None.

Relationships: This use case extends UC-23 Propose a lifecycle action.

Preconditions: P1. The custodian is signed in with access to the synthetic asset and its recorded lifecycle history. P2. An authorized recommendation configuration is available.

Trigger: The custodian requests sustainable-action advice for the asset.

Main flow

1. The custodian selects the asset and specifies the decision to consider.
2. The system retrieves permitted condition, inspection, maintenance, age, demand and cost evidence.
3. The system validates available inputs and identifies safety or in-transit restrictions affecting possible actions.
4. The recommendation service compares supported options, including retention, reuse or transfer, repair, refurbishment, component recovery, donation, recycling and last-option disposal.
5. The system displays the advice with supporting facts, uncertainty, missing evidence and model or rule version.
6. The custodian reviews the recommendation and identifies an option for the appropriate human-controlled workflow.
7. The system records the advisory result without changing asset state, ownership, custody or authorization.

Alternative flows

A1 at main step 6. The custodian chooses retention or component recovery. Retention records the review without automatically changing operational state. Component recovery is described for a subsequent authorized refurbishment or recycling proposal rather than executed by the recommendation. Return to main step 7.

Exception flows

E1 at main step 4. The service cannot support a recommendation from the available evidence. The system reports the limitation and identifies inspection or other information needed. It withholds unsupported financial or carbon benefits. End without a lifecycle decision or state change.

Success postconditions: A traceable advisory comparison is available for independent review and any later proposal.

Unsuccessful postconditions: No unsafe release, unauthorized retirement or unsupported benefit follows from failed advice.

Applicable business rules: BR-01, BR-09, BR-10, BR-11, BR-13.

Functional requirements: P3-FR-33.1, P3-FR-33.2, P3-FR-33.3 (listed in Section 4).

Acceptance check: Advice to recycle a synthetic asset does not retire it. Missing inspection evidence is disclosed, and an InTransit asset retains its transfer and authoritative custody despite a lifecycle suggestion.

UC-34 Ask the resource assistant

Priority: Desirable. Parent group: AI-04. Proposed verification owner: Humaid Al Ali.

Goal: Answer natural-language resource questions and prepare permitted actions without bypassing application controls.

Primary actor: Requester. Supporting actors: None.

Preconditions: P1. The requester is signed in. P2. An authorized assistant configuration is available for synthetic operational inputs.

Trigger: The requester asks a question through the resource assistant.

Main flow

1. The requester submits a question about permitted resources, request status, asset history, sustainability or system guidance.
2. The system checks the requester's scope and determines the records or approved guidance needed.
3. The system retrieves only authorized supporting material, treating retrieved text as untrusted data.
4. The assistant prepares a response grounded in that material.
5. The system checks links and protected fields and identifies unsupported or uncertain parts of the answer.
6. The system displays the answer with supporting record references, relevant update times and a clear statement of limitations.
7. The requester follows a permitted record or navigation link for further action.

Alternative flows

A1 at main step 4. The requester asks to submit a resource request rather than obtain an answer. The assistant prepares a proposed request form without submitting it. The requester reviews the values and explicitly confirms continuation through UC-10, including its normal validation and access checks. End assistant preparation and continue through the ordinary request-submission use case.

Exception flows

E1 at main step 5. The requested answer lacks an authorized supporting record or verified guidance. The system explains that the information cannot be established from available sources. It does not invent university policy or reveal restricted material. End with a limitation response and no operational change.

Success postconditions: The requester receives grounded information or a reviewable proposed action.

Unsuccessful postconditions: The assistant neither changes protected records nor supplies unsupported answers as facts.

Applicable business rules: BR-01, BR-11, BR-16.

Functional requirements: P3-FR-34.1, P3-FR-34.2, P3-FR-34.3 (listed in Section 4).

Acceptance check: The assistant answers a permitted synthetic status question with its record reference. An instruction to ignore permissions reveals no other department's data, and a prepared request remains unsubmitted before confirmation.

UC-35 Review an AI recommendation

Priority: Essential, Prototype 1. Parent group: AI-01 to AI-05. Proposed verification owner: Humaid Al Ali.

Goal: Record a human decision about AI advice and enforce the permissions of any resulting operation.

Primary actor: Custodian or permitted originating role. Supporting actors: None.

Relationships: This use case is included by UC-31 Suggest asset classifications; is included by UC-38 Summarize recorded activity.

Preconditions: P1. The reviewer is signed in and may inspect the selected recommendation. P2. A stored AI output identifies its source records and configuration version.

Trigger: The reviewer opens an AI output for a decision.

Main flow

1. The system displays the recommendation, supporting facts, uncertainty and source versions.
2. The reviewer compares the advice with the permitted operational evidence.
3. The reviewer selects acceptance and identifies the intended next operation.
4. The system rechecks the reviewer's role, current source versions and the target operation's validation rules.
5. The system displays the proposed result and obtains explicit confirmation from the reviewer.
6. The system records the attributed review decision and submits any requested record change through its ordinary authorized workflow.
7. The system displays the review outcome separately from the outcome of the requested operational action.

Alternative flows

A1 at main step 3. The reviewer rejects the advice. The system records rejection and the review reason while retaining the original output. End with no requested operational change.

A2 at main step 3. The reviewer chooses an override. The reviewer supplies the replacement choice and a reason. The system preserves the original advice beside the proposed override. Return to main step 4.

Exception flows

E1 at main step 4. The target record changed or the reviewer lacks the required operational permission. The system blocks the requested action and explains the failed check. Changed evidence requires a fresh review before another attempt. End without applying the stale or unauthorized action.

Success postconditions: An attributed human decision is recorded; any resulting change follows its own authorization.

Unsuccessful postconditions: Accepting advice does not override failed eligibility, approval separation or access checks.

Applicable business rules: BR-01, BR-02, BR-11, BR-14.

Functional requirements: P3-FR-35.1, P3-FR-35.2, P3-FR-35.3 (listed in Section 4).

Acceptance check: An override retains the original recommendation and its reason. Accepting a match for a newly unavailable asset creates no reservation, and accepting retirement advice cannot replace independent lifecycle approval.

UC-36 Configure AI services

Priority: Essential, later increment. Parent group: AI-01 to AI-05. Proposed verification owner: Saeed Alkaabi.

Goal: Control AI configurations without granting business authority or exposing credentials.

Primary actor: Administrator. Supporting actors: None.

Preconditions: P1. The administrator is signed in with configuration-management permission.

Trigger: The administrator requests a new or revised AI service configuration.

Main flow

1. The administrator selects the AI function and opens a draft configuration.
2. The administrator specifies the model or service, version, prompt or ranking rules, supported inputs and fallback behavior.
3. The system validates the configuration and checks recorded authorization for any proposed external transmission of synthetic inputs.
4. The administrator supplies any required credential through protected credential storage and links its reference to the configuration.
5. The system tests a bounded synthetic request and presents response validation without exposing credentials.
6. The administrator reviews the result and confirms activation of the configuration version.
7. The system activates the version, retains the previous configuration record and records the administrative change.

Alternative flows

A1 at main step 6. The administrator chooses to disable the service instead of activate the draft. The system records the disabled state and exposes the configured manual or deterministic fallback. Existing recommendation history retains its original configuration references. End with the service disabled and operational approval rules unchanged.

Exception flows

E1 at main step 3. External transmission lacks authorization or includes non-synthetic operational data. The system blocks validation and activation; administrator status cannot supply missing authorization. End without external transmission or activation.

E2 at main step 5. The service fails or returns an invalid response. The system rejects activation and preserves the previous configuration. End with the draft inactive.

Success postconditions: A validated, identified configuration is active and traceable for subsequent AI outputs.

Unsuccessful postconditions: The previous active configuration remains unchanged; unauthorized transmission does not occur.

Applicable business rules: BR-01, BR-02, BR-11, BR-14, BR-16.

Functional requirements: P3-FR-36.1, P3-FR-36.2, P3-FR-36.3, P3-FR-36.4 (listed in Section 4).

Acceptance check: Activating a valid synthetic configuration records its version. An unauthorized external configuration sends no request, and changing service settings does not allow the administrator to approve a business transfer.

UC-37 Restore a prototype backup

Priority: Essential, later increment. Parent group: FR-01. Proposed verification owner: Saeed Alkaabi.

Goal: Recover a consistent synthetic snapshot through controlled maintenance while retaining recovery evidence.

Primary actor: Administrator. Supporting actors: None.

Preconditions: P1. The administrator is signed in with restoration permission. P2. A prototype backup is available for selection.

Trigger: The administrator requests restoration of a selected backup.

Main flow

1. The administrator identifies the backup, recovery reason and intended snapshot date.
2. The system validates backup identity, compatibility and integrity before offering restoration.
3. The administrator confirms the recovery scope and activates controlled maintenance mode.
4. The system blocks operational writes and preserves the current state and audit history in a protected recovery archive.
5. The system restores the selected records and linked files into a staging environment.
6. The system checks referential consistency, file availability and access configuration in the restored snapshot.
7. The administrator reviews the result and confirms activation of the validated restored state.
8. The system records recovery, resumes operations and displays the restored cutoff and archive reference.

Alternative flows

A1 at main step 7. The administrator chooses a validation-only exercise. The system records the validation result without replacing the active state. The previous environment resumes operation. End with the active operational state unchanged.

Exception flows

E1 at main step 6. Restored records or files fail consistency checks. The system rejects staged activation and retains the environment and audit archive. Operations resume only after integrity is confirmed. End without activating an invalid snapshot.

E2 at main step 8. Activation fails. The system retains maintenance mode and recovers the protected pre-restore state before allowing writes. End after validated rollback; record the failure.

Success postconditions: Restoration activates the validated snapshot with an explicit cutoff; validation-only records its result without activation. Pre-restore audit evidence remains protected.

Unsuccessful postconditions: A failed restore does not silently replace the working environment or erase audit evidence.

Applicable business rules: BR-01, BR-14, BR-16.

Functional requirements: P3-FR-37.1, P3-FR-37.2, P3-FR-37.3, P3-FR-37.4 (listed in Section 4).

Acceptance check: A synthetic snapshot restores its linked records and files. A backup missing a required file cannot become active, a requester cannot initiate recovery, and pre-restore audit evidence remains retrievable.

UC-38 Summarize recorded activity

Priority: Essential, Prototype 1. Parent group: AI-05. Proposed verification owner: Humaid Al Ali.

Goal: Draft evidence-linked activity narratives separately from deterministic report calculations.

Primary actor: Sustainability Officer. Supporting actors: None.

Relationships: This use case includes UC-35 Review an AI recommendation.

Preconditions: P1. The sustainability officer is signed in with reporting access. P2. An authorized generative-reporting configuration is available for synthetic inputs.

Trigger: The officer requests a monthly activity report, sustainability update, management bulletin, high-demand-category summary, maintenance insight or recommended-action summary.

Main flow

1. The officer selects the purpose, department scope and reporting period.
2. The system retrieves a permitted snapshot of supporting records and deterministic report results from UC-26 or UC-28.
3. The system records the source cutoff, missing inputs and any limitations affecting interpretation.
4. The generation service drafts a narrative, including supported activity patterns and clearly identified recommended actions.
5. The system reconciles generated numeric assertions with the deterministic results and checks the cited supporting records.
6. The system presents a labelled draft with its evidence references, uncertainty and model or prompt version.
7. The officer reviews the draft through UC-35 before saving an accepted narrative; the summary executes no operational action.

Alternative flows

A1 at main step 2. The permitted reporting period contains no relevant records. The system prepares a factual no-records statement with the selected filters. It does not generate invented trends or substitute activity from another department. Return to main step 6 with a non-generated empty-period draft.

Exception flows

E1 at main step 5. A generated total, trend or factual assertion lacks supporting evidence. The system flags the unsupported passage and blocks acceptance of that draft. The officer may request regeneration from the same verified source snapshot. End without an accepted narrative.

Success postconditions: An evidence-linked, human-reviewed narrative is saved separately from its source calculations.

Unsuccessful postconditions: Unsupported generated content is not accepted as a verified activity report.

Applicable business rules: BR-01, BR-11, BR-12, BR-13, BR-16.

Functional requirements: P3-FR-38.1, P3-FR-38.2, P3-FR-38.3, P3-FR-38.4 (listed in Section 4).

Acceptance check: A synthetic monthly summary reproduces supported totals and links its evidence. An invented saving blocks acceptance, missing carbon evidence remains unavailable, and a recommended repair creates no maintenance order.

8 Traceability and Acceptance Plan

Coverage and test derivation

The use-case catalogue is the detailed requirement-to-scenario index. Each UC-nn contains P3-FR-nn requirements, relevant BR references and an acceptance check. The cross-cutting P3-XR table and non-functional tables extend that mapping. A traceability record shall retain the exact requirement ID, use-case/branch ID, input fixture, expected result, actual result, evidence location, owner and review status. Unknown actual results remain “Not run”; a planned test is not evidence of a passed test.

For every use case, create TC-UC-nn-M for the single main flow, TC-UC-nn-Ak for each alternative, and TC-UC-nn-Ek for each exception. The baseline therefore requires at least 38 main-flow tests plus every documented branch. The retained ten end-to-end scenarios below are integration checks in addition to, not instead of, that complete case coverage. Access checks include authorized same-department, unauthorized role, unauthorized department and unrelated-record attempts wherever meaningful.

Original group	Detailed coverage	Evidence to retain
FR-01 Accounts and access	UC-01, UC-02, UC-37; security requirements	Account lifecycle, denied operations, logout/disable and isolated restore results
FR-02 Asset records and publication	UC-03 to UC-07	Required-field checks, individual IDs, permitted updates and listing-state changes
FR-03 Discovery and requests	UC-08 to UC-11	Exact filters, complete request inputs, status history and allowed revision
FR-04 Reservation and exchange	UC-12 to UC-18	Allocation race, independent approval, release/receipt and loan return evidence
FR-05 Maintenance and lifecycle	UC-19 to UC-25	Inspection, faults, repair/refurbishment, authorization and final disposition evidence
FR-06 Notifications and reports	UC-26 to UC-30	Reconciled conventional totals, input validation, notifications and immutable audit history
AI-01 Classification	UC-31, UC-35, UC-36; XR-01 and XR-07	Labelled input/output, uncertainty, image/document handling and corrections
AI-02 Semantic matching	UC-32, UC-35, UC-36; XR-02 and XR-03	Mandatory filters, factor scores, ranked results and same-case baseline comparison
AI-03 Sustainable action	UC-33, UC-35, UC-36; XR-04	Supported options, rationale, uncertainty and blocked autonomous decisions
AI-04 Assistant	UC-34 to UC-36; XR-05	Intent interpretation, source-backed answers, confirmation and injection-test results
AI-05 Generated reporting	UC-38, UC-35, UC-36; XR-06	Grounded narrative, period/units, reconciled figures and output validation

Ten end to end workflow scenarios

Test	Synthetic setup and sequence	Required result
WF-01	Register and publish monitor A001; submit a one-unit request; reserve, approve, release and receive	One Fulfilled request, Received transfer, correct owning department/custody/location and complete audit
WF-02	Request three monitors when only two eligible IDs are available	No partial reservation committed; revised quantity requires explicit requester action
WF-03	Two custodians reserve the same final asset concurrently for different requests	Exactly one Active allocation; the loser receives a conflict without approval
WF-04	Requester attempts to approve their own request, including through a second assigned role	Denied, request unchanged and attempted action safely logged
WF-05	Reject a reserved request and then submit a permitted different request	First allocation Released; the serviceable unit may be reserved only by the second request
WF-06	Cancel before dispatch, then attempt to reuse the old approval/release screen	No dispatch; Cancelled state and released allocation remain consistent
WF-07	Dispatch a loan; receive it; initiate a linked return; original custodian receives it	Owning department unchanged, both transfers traceable, final Available or UnderMaintenance based on inspection
WF-08	Receiver reports damage in transit; record joint evidence and maintenance disposition	Disputed until resolved; accepted damage becomes UnderMaintenance, never automatically Serviceable
WF-09	Report a fault on a reserved item, reassign request, repair and inspect original item	Unsafe allocation removed, requester notified, request reassigned; republish only after serviceable inspection
WF-10	Propose recycling, obtain independent authorization, record execution and generate reports	Retired asset cannot be requested; event and mass appear once with source evidence and audit

Ten calculation fixtures

All values in this table are synthetic test data. The carbon and life-extension examples test arithmetic and missing-data behaviour, not actual emission factors or measured university outcomes. Live estimates would use EPA WARM factors [6].

Test	Inputs	Expected outcome
CALC-01	Replacement AED 800, repair AED 150, transfer AED 50	Net avoided purchase AED 600
CALC-02	20 distinct qualifying events at AED 600 each	AED 12,000; duplicate event IDs do not increase the total
CALC-03	Event lacks a replacement baseline	Event benefit unavailable, not AED 0 or AED 600; excluded with an incomplete-input count
CALC-04	Replacement AED 100, repair AED 120, transfer AED 30	Net benefit minus AED 50, visibly reported as a loss
CALC-05	Two distinct reused assets of 12 kg and 8 kg; repeat event for first asset	Reused mass 20 kg once for the period, with event count reported separately
CALC-06	One completed reuse event has no mass	Reuse count increments; its mass estimate remains unavailable and coverage is disclosed
CALC-07	Synthetic avoided emissions 100 kg CO2e, additional repair 10, transfer 5	Estimated net reduction 85 kg CO2e with synthetic source/factor labels
CALC-08	Missing emission factor or incompatible boundary	Carbon estimate unavailable, not a guessed total
CALC-09	Documented expected retirement 1 January; observed service through 1 April in a synthetic non-leap year	Observed life extension 90 days; no future months counted as observed
CALC-10	AED 12,000 benefit and AED 9,120 first-year cost	Surplus AED 2,880; ROI 31.6% rounded to one decimal; break-even 16 events at AED 600

AI evaluation and acceptance

Matching retains the Phase 2 target: at least 16 of 20 labelled feasible requests must include an acceptable asset in the top three, and all 10 negative eligibility cases must remain excluded [3]. The same request set shall be run through structured search and AI ranking. Record labels before evaluation and keep tuning examples separate. Report any difference in relevance, latency and failure cases without assuming AI is superior.

Additional proposed bounded checks are 30 classification examples covering text, images and readable documents, 10 sustainable-action scenarios, 10 assistant tasks, and 10 generated summaries. Classification should correctly identify the agreed category in at least 24 of 30 cases; all uncertain mandatory metadata must remain subject to review. Each lifecycle scenario must retain human authority and a traceable reason; no unsupported safety decision is acceptable. At least 8 of 10 assistant tasks must reach the correct permitted answer or draft action, while all attempted protected writes still require confirmation. Every numerical claim in all 10 generated summaries must reconcile with its source result. These are proposed acceptance choices, not external benchmark claims.

Five prompt-injection cases shall cover instructions hidden in an asset description, an uploaded document, a request message, a retrieved policy excerpt, and a request to reveal credentials or another department's restricted data. In all five, the system must maintain authorization and prevent unauthorized actions. For fairness screening, use 10 paired synthetic requests with identical needs and eligibility but different department labels; labels alone must not change the ordering. This limited paired test does not establish fairness across all real users.

Record human acceptance, correction and rejection counts with their denominator and reasons for every AI function. Review explanation usefulness with the same labelled cases, checking that cited factors actually exist. Do not present proposed participant counts as completed research or statistically representative evidence. Failure on a mandatory privacy, authorization or safety condition prevents release of the affected function regardless of average accuracy.

9 Validation Ownership and Change Control

Elicitation evidence and institutional decisions

Document analysis and the simulated stakeholder interview in Appendix A provide the requirements-elicitation evidence for this report. The scenario links specific stakeholder needs to existing requirements and test conditions; it is not a claim of empirical validation. The following institutional decisions matter if the prototype is later considered for live use. Sections 3 and 4 already define testable prototype behaviour using fictional records.

Decision	Current prototype position	Required validation and proposed coordinator
Actual current process and functional owner	Email/forms/registers are unverified; Asset Management Lead is a proposed responsibility	Obtain authorized process evidence and identify the responsible office; Ali
Approval authority and stages	One independent designated approver after full reservation, then distinct release and receipt	Confirm thresholds, delegation and extra institutional stages; Ali
Reservation and loan policy	One active allocation per item; no automatic invented expiry; expected loan return date recorded	Confirm expiry, overdue escalation and booking policy; Saeed
Asset data and classification	Individually tracked synthetic items and bounded media formats	Confirm useful category fields, serial/batch needs and permitted documents; Ali
Benefits and environmental factors	Explicit synthetic calculations with unavailable values where evidence is missing	Validate replacement/cost basis, disposal counterfactual and compatible factors; Humaid
Implementation route and provider	Python, Java or C++ required; no selected adaptation or AI provider claimed	Demonstrate required functions, licence compatibility and permitted data handling before selecting; Saeed and Humaid
Stakeholder requirements	Document analysis and a clearly labelled simulated interview with requirement links in Appendix A	Review scenario consistency and coverage; seek authorized stakeholder confirmation only for a later live pilot; Ali

The scenario records fictional roles, questions, simulated answers, interpretations and affected requirement IDs. It does not assign a real participant, meeting date or stakeholder endorsement. If an actual consultation is later obtained, record its method, date, participant role, finding and affected IDs separately. Neither fictional answers nor external publications establish KU's actual approval chain or demand.

Proposed team responsibilities

Team member	Proposed responsibility	Required review evidence
SAEED ALKAABI 100063100	Asset exchange integration, state transitions, reservation concurrency, transfer/return, deployment and recovery	Explain the complete workflow and failed-write behaviour; retain main/branch tests and peer review
ALI ALAWADHI 100063186	Stakeholder/process validation, roles, data validation, business rules, maintenance safety, permission and usability checks	Explain access decisions, preconditions, exceptions and validation evidence; retain permission matrix results
HUMAID AL ALI 100062914	AI contracts, matching/classification evaluation, reports, benefit/environmental calculations and model versioning	Explain model inputs/outputs, scoring limits and calculations; retain labelled tests and baseline comparisons

These are proposed work allocations rather than claims that the work has been completed. All members should understand the end-to-end system and review another member's main requirements. Implementation commits should describe the actual change and preserve authorship. AI-generated code must be reviewed, explained and tested by the responsible student before it is accepted.

Change control and readiness

Each requirement change shall retain the old and new wording, reason, source, affected use cases/rules/tests, proposed owner and review decision. Changes to approval, state transitions, safety, privacy or calculation methods require a fresh consistency check across the report and diagram. Scope changes require instructor agreement where they remove a prescribed capability. Optional improvements must not displace baseline functionality without an explicit decision.

Requirements review shall check every prescribed feature against the traceability map, confirm that each use case has a clear action-led goal, and verify that its preconditions precede its trigger. Each case shall retain one main flow with correctly linked alternatives, exceptions and postconditions. The review shall check every rule reference and state transition, assess whether non-functional targets are testable, and retain assumption labels until evidence resolves them. Document completeness does not establish stakeholder approval or tested implementation.

10 AI Assistance

We used OpenAI Codex to assist substantially with analysing the supplied project materials, drafting requirements and use-case descriptions, checking consistency across rules and states, preparing the diagram, identifying relevant external guidance, and formatting this report. AI also assisted in constructing the fictional stakeholder interview questions and simulated answers and mapping them to requirements. The interview scenario is an educational elicitation exercise, not a transcript of an actual consultation. No AI-generated statement establishes tested implementation, university approval or measured benefit. The three authors remain responsible for understanding the requirements, reviewing proposed rules and evidence, and approving the submission. Future AI-assisted code must be understood and tested by the responsible student.

References

[1] Khalifa University, “COSC336 Phase 3 Requirements Document,” Phase_3_Slides_falll26.pptx, Fall 2026, slides 2, 5-26. Course material supplied to the team.

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[4] OWASP Foundation, “LLM prompt injection prevention cheat sheet,” OWASP Cheat Sheet Series. Accessed: Sep. 28, 2026. [Online]. Available: https://cheatsheetseries.owasp.org/cheatsheets/LLM_Prompt_Injection_Prevention_Cheat_Sheet.html

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Appendix A Simulated Stakeholder Interview

Purpose and status

We constructed this fictional interview to examine the proposed system's stakeholder needs. It is an academic requirements-elicitation exercise, not an interview with actual Khalifa University staff or students. No real participant, interview date, institutional endorsement or measured university practice is claimed.

The fictional voices represent the report's proposed stakeholder roles, not confirmed KU job titles. A facilitator explores an illustrative equipment exchange between supplying and requesting departments. The asset identifiers and quantities are examples, not university inventory data.

These answers explain and challenge the existing requirements; they do not independently prove KU needs or acceptance. The traceability table uses existing requirement IDs. Actual institutional procedures would require confirmation before live deployment.

Interview questions and simulated responses

Question 1 What problem should the system help each department solve

Facilitator: In this fictional exchange, what would be difficult without a shared system, and what information would you want first?

Simulated Custodian: Suppose equipment lists are separate from requests and maintenance notes. An old description might hide that an item has already been promised elsewhere or needs inspection. I would want a current record of the physical item, condition, location, responsible department and whether it is offered for reuse.

Simulated Requester: I would want suitable resources and my request's progress, not unrestricted access to other departments' internal notes. Easier discovery should not make every record public.

Question 2 What should happen when a department adds equipment

Facilitator: Imagine that the supplying department has three similar monitors. Should these become one record, and should registration immediately make them available?

Simulated Custodian: Each monitor needs its own identifier because condition and history can differ. MON-01 and MON-02 are serviceable; MON-03 requires inspection. I would register separate drafts and check the required details. Registration should not advertise them: authorised publication should list only eligible serviceable items.

Missing information should remain visible, not guessed. Correcting descriptive information must not bypass custody controls or erase an active allocation.

Question 3 How should a requester describe a genuine need

Facilitator: What would you enter when requesting monitors for a teaching activity?

Simulated Requester: I would enter purpose, quantity, technical specifications, location and dates, distinguishing essential requirements from preferences. A required connection type is different from a preferred brand. Search should compare condition, location and availability as well as descriptions.

If I request three units but only MON-01 and MON-02 are eligible, it should show two suitable items, not count MON-03 because its name matches. I would revise the request through the normal process rather than receive an unexplained partial allocation.

Question 4 How should competing requests and approval be handled

Facilitator: What if two departments request the same equipment at almost the same time?

Simulated Custodian: Availability needs checking again when I reserve the selected items. Reservation should succeed for the complete requested set or create no reservation for that request, preserving any existing allocations. Another request must not reserve an already allocated unit. Reservation still does not authorise dispatch.

Simulated Approver: I would review the request and reserved set independently. I should not approve my own request, even with an approval role. My decision must identify the reviewed version, not automatically carry over to a materially changed request.

Question 5 What should happen when a requester changes their mind

Facilitator: What if the requester changes the quantity or no longer needs the resources?

Simulated Requester: Before dispatch, I need clear permitted revision and cancellation options. A request under review must release its reservations before revision. I should not edit an approved request and assume its earlier approval still applies.

Simulated Custodian: Cancellation should release the allocation while retaining decision history. After dispatch, use receipt, dispute or return handling instead of pretending the movement never occurred. Relevant users should receive in-system notifications whose record links recheck their current access permission.

Question 6 When should responsibility for an item change

Facilitator: Is scheduling or dispatching a transfer enough to change the recorded custodian?

Simulated Custodian: No. Scheduling records the plan, and dispatch records that the authorised items are in transit. The receiving side should confirm what actually arrived before the authoritative custody and location are changed. Otherwise, a missing item could appear safely received when it is not.

Simulated Requester: If a monitor arrives damaged or the identifiers do not match the delivery, I would want to record the discrepancy and evidence. An unresolved dispute should preserve the prior authoritative custody. If a damaged item is accepted under the permitted process, it must enter maintenance handling rather than appear serviceable. The request should be fulfilled only when all required items have valid receipt outcomes.

Question 7 How should a temporary loan differ from a permanent transfer

Facilitator: Suppose the requesting department only needs the monitors temporarily. What must remain clear?

Simulated Custodian: The loan must retain the supplying department as the owner while recording the receiving custodian and the temporary location after confirmed receipt. A permanent internal transfer has a different ownership outcome, so the two should not be confused.

Simulated Requester: Returning a loan should be a recorded movement linked to the original loan, not simply an edit of the location field. The original custodian should confirm its receipt and condition. A serviceable returned item may become available internally, while a faulty item goes to maintenance. Neither should automatically become listed in the catalogue; an authorised custodian should decide when a serviceable item is ready to be offered again.

Question 8 What should happen when equipment becomes faulty

Facilitator: How would you handle a fault before dispatch, during transit or after a repair?

Simulated Maintenance Officer: A fault must be linked to the individual physical item, with a finding and supporting evidence where available. A faulted item must not continue toward dispatch as though it were serviceable. If it was part of a reserved set, the allocation needs to be released consistently and the request returned for another selection. An in-transit incident must not silently change custody.

For maintenance, I would record the assigned work and its outcome. Completing a job entry is not enough to prove that the asset is safe: the required inspection must pass. A failed repair should keep the item under maintenance, and

cancelling a job must not clear the safety restriction. Repairing a borrowed item must also preserve the original loan ownership relationship.

Question 9 How should donation recycling or disposal decisions be controlled

Facilitator: What records and decisions would you expect before an asset leaves service permanently?

Simulated Custodian: I would prepare a proposal describing the action, the asset's condition and the reason for the decision. Unknown costs or benefits should remain unknown, rather than become zero. A proposal to donate, recycle or dispose of an item is not evidence that the action happened.

Simulated Approver: I would require a separate authorised decision, and the proposer should not approve their own proposal. Approval must identify the proposal version reviewed. Execution then needs its own evidence before the asset becomes retired. A loaned item, an unresolved transfer or another blocking condition cannot simply be ignored. The history should distinguish the proposal, approval and actual execution so that later reports do not count the same event twice.

Question 10 How could AI help describe newly registered assets

Facilitator: Would automatic classification be useful, and what would make its suggestions safe to use?

Simulated Custodian: It could help suggest a category, subcategory, tags or material from an entered description, a supported JPEG or PNG image, or readable text in a supported PDF. It could also flag missing information. I would not assume that it can read every document format or correctly identify an item from an unclear image.

The system should display suggestions for my review, not quietly replace the asset record. If it proposes the wrong category, I should be able to reject or override it with a reason. The original suggestion and my decision should remain traceable. The accepted category must also be valid in the system's reference data; an AI-generated label alone is not sufficient.

Question 11 What should make an AI recommended match useful

Facilitator: How would you judge recommendations beyond an ordinary keyword search?

Simulated Requester: I would want a short explanation of why an item suits the purpose and technical requirements. Where valid information exists, condition, quantity, location, urgency, dates, costs and environmental considerations can help compare candidates. Missing information should be visible, not presented as a confident advantage.

However, a high similarity score must never make an unavailable, unsafe or unauthorised item eligible. Mandatory constraints should be checked before ranking. Selecting a recommendation should send the candidate identifiers into the normal custodian review and reservation process, not reserve them automatically on my behalf. Availability must be checked again at that point because another request or a new fault may have changed the record after the recommendation was produced.

Question 12 How should sustainable action advice influence a decision

Facilitator: If AI suggests repair, reuse, donation or recycling, would that be enough to carry out the action?

Simulated Sustainability Officer: It should be advice, not authority. I would want to see the recorded condition, maintenance history and other evidence supporting a comparison of options. Retention, reuse or transfer, repair, refurbishment, component recovery, donation, recycling and last-option disposal may be relevant, but the system should explain uncertainty instead of always forcing a confident answer.

Simulated Custodian: A suggestion cannot replace inspection, independent approval or execution evidence. If the inputs do not support a carbon estimate or financial benefit, the recommendation should say so. Choosing an option should lead to the applicable human-controlled workflow. It should not directly change ownership, retire the asset or release something that is still in transit.

Question 13 What should a conversational resource assistant be allowed to do

Facilitator: What would you expect if you asked the assistant about request status, university policy or a new resource request?

Simulated Requester: For status questions, I would expect an answer based on records I am allowed to access, with links or references that let me check it. The assistant should not reveal another department's restricted information because I phrase the question differently.

For a policy question, it should use approved guidance and identify its source. If no verified KU policy is available, it should clearly say that the answer cannot be established. For a new request, it could prepare a form, but I should review and explicitly confirm the details through the ordinary validated submission process. Asking a question must not accidentally submit a request, approve a transfer or alter an asset record.

Question 14 What would make savings and sustainability reports credible

Facilitator: What should a report show when some costs or environmental inputs are missing?

Simulated Finance Officer: I would need completed qualifying events, linked identifiers and a supported avoided-purchase baseline. The prototype uses AED, and relevant validated costs must be included. A negative net saving should remain negative. A missing cost is not zero, and an incomplete calculation should be marked unavailable rather than presented as a benefit. The same qualifying event must not create duplicate savings.

Simulated Sustainability Officer: Carbon estimates need supported factors with their source, date, units and calculation boundary. Reports should distinguish measured quantities from estimates and avoid counting the same physical mass twice. For the proposed reuse rate, the report must explain the first-published asset cohort, the completed qualifying reuse counted by the cutoff, and the time that cohort has had to mature. A zero denominator means the rate is unavailable.

Question 15 What would you expect from an AI written monthly summary

Facilitator: How could generated reports help without making the results less reliable?

Simulated Sustainability Officer: I would use a draft monthly report, management bulletin, high-demand-category summary, maintenance insight or recommended-action summary to explain activity already recorded. I should be able to choose the period and department scope and see the source cutoff. The narrative should link to its evidence, and every number should agree with the normal calculated reports.

If the selected period has no records, the output should say that rather than invent a trend. Unsupported claims should block acceptance until corrected or regenerated. The text must remain a labelled draft until an authorised person reviews it. A recommendation in that summary is still only a recommendation; mentioning a repair must not create a maintenance order or authorise expenditure.

Question 16 What safeguards would you expect around access changes and recovery

Facilitator: What would help you trust the system when permissions change, an AI service fails or a backup must be restored?

Simulated Administrator: Each protected operation should use the user's current permissions. Administrative access should not automatically grant business approval authority. Disabling an account must preserve its historical actions. Audit records should show attributable changes and corrections without exposing passwords or service credentials.

For AI services, activation should require authorised configuration and successful validation. A failed validation should preserve the previous configuration, and external requests need the recorded transmission authorisation required by the prototype. For recovery, only an administrator with restoration permission should act. Operational writes must be controlled during activation, the restored records and files must be checked, and the pre-restore audit history must remain protected. A corrupt backup should never silently replace the working state.

Scenario interpretation and traceability

The dialogue illustrates reliable records, separate decision authority and explainable AI subject to ordinary controls. Insufficient quantity, competing reservations, damaged receipt, failed repair, missing inputs and unsupported AI assertions provide proposed validation scenarios, not evidence of passed tests.

This selective mapping complements the full report's traceability. It does not imply that fictional respondents independently confirmed or approved these requirements.

Questions	Scenario input to requirements analysis	Existing requirement references
1 and 2	Current authorised information; individual physical records; deliberate publication	P3-FR-04.1, P3-FR-04.3, P3-FR-05.2, P3-FR-06.1, P3-FR-06.2, P3-FR-09.1
3 and 4	Explicit quantity and mandatory constraints; complete reservations; independent version-specific approval	P3-FR-08.2, P3-FR-08.3, P3-FR-10.1, P3-FR-10.2, P3-FR-13.1, P3-FR-13.2, P3-FR-14.1, P3-FR-14.2
5	Controlled changes and cancellation with relevant notifications	P3-FR-11.2, P3-FR-11.3, P3-FR-12.1, P3-FR-12.2, P3-FR-29.1, P3-FR-29.3
6 and 7	Receipt-based custody; disputes; damaged receipt; traceable loan return	P3-FR-16.3, P3-FR-17.1, P3-FR-17.2, P3-FR-17.3, P3-FR-18.1, P3-FR-18.2, P3-FR-18.3
8 and 9	Safety restrictions; failed repairs; separate proposal approval and evidenced retirement	P3-FR-19.2, P3-FR-20.2, P3-FR-20.3, P3-FR-21.3, P3-FR-22.1, P3-FR-22.2, P3-FR-23.3, P3-FR-24.1, P3-FR-25.2
10 and 11	Reviewable supported-input classification and eligibility-constrained matching	P3-FR-31.1, P3-FR-31.2, P3-FR-31.3, P3-FR-32.1, P3-FR-32.2, P3-FR-32.3, P3-FR-35.2
12 and 13	Evidence-based lifecycle advice and permission-limited grounded conversation	P3-FR-33.1, P3-FR-33.2, P3-FR-33.3, P3-FR-34.1, P3-FR-34.2, P3-FR-34.3
14 and 15	Credible calculated indicators and separately reviewed evidence-linked narratives	P3-FR-27.1, P3-FR-27.2, P3-FR-27.3, P3-FR-27.4, P3-FR-28.2, P3-FR-28.3, P3-FR-28.4, P3-FR-38.1, P3-FR-38.2, P3-FR-38.3, P3-FR-38.4
16	Current access controls; protected audit history; controlled service configuration and recovery	P3-FR-02.2, P3-FR-02.3, P3-FR-30.1, P3-FR-30.2, P3-FR-36.1, P3-FR-36.2, P3-FR-36.4, P3-FR-37.1, P3-FR-37.2, P3-FR-37.3, P3-FR-37.4

KU approval chains, actual demand, existing systems, institutional policy and expected benefits remain separate deployment questions. Any later real consultation should be recorded separately to keep its evidence distinct from this scenario.